Journal of Algebra 704 (2026) 139–164

Contents lists available at ScienceDirect

Journal of Algebra

journal homepage: www.elsevier.com/locate/jalgebra

Research Paper

Properties of quasi-projective dimension over abelian categories

Hongxing Chen [a,*], Xiaohu Chen [b], Mengge Liu [b]

[a] *School of Mathematical Sciences & Academy for Multidisciplinary Studies, Capital Normal University, 100048 Beijing, PR China*
[b] *School of Mathematical Sciences, Capital Normal University, 100048 Beijing, PR China*

ARTICLE INFO

Article history:
Received 26 September 2025
Available online 12 May 2026
Communicated by Volodymyr Mazorchuk

MSC:
primary 16E10, 18G20, 16G10
secondary 18E10, 16E35

Keywords:
Quasi-global dimension
Quasi-projective dimension
Nakayama algebra
Self-orthogonal module

ABSTRACT

Quasi-projective dimension was introduced by Gheibi, Jorgensen and Takahashi to generalize the Auslander-Buchsbaum formula and the depth formula in commutative algebra. In this paper, we establish some basic properties of quasi-projective dimension of objects in abelian categories. Analogous to global dimension of rings, we also introduce the concept of quasi-global dimension for left Noetherian rings, and then compare quasi-global dimension with global dimension for a class of Nakayama algebras. This provides new examples of finite-dimensional algebras with finite quasi-global dimension but infinite global dimension.

1. Introduction

In the representation theory of algebras and homological algebra, projective dimension of objects in abelian categories plays a very important role. In [8], Gheibi,

* Corresponding author.
E-mail addresses: chenhx@cnu.edu.cn (H. Chen), xiaohu.chen@cnu.edu.cn (X. Chen), 1602567185@qq.com (M. Liu).

https://doi.org/10.1016/j.jalgebra.2026.04.046

Jorgensen and Takahashi introduced a generalization of projective dimension, called *quasi-projective dimension*. This new dimension not only supplies a general framework to establish the Auslander–Buchsbaum formula and the depth formula in commutative algebra for modules of finite quasi-projective dimension (see [9]), but also gives an equivalent characterization of equipresented, local, complete intersection rings in terms of finite quasi-projective dimension (see [3]). Dual to quasi-projective dimension, quasi-injective dimension of modules over rings was introduced to provide extensions of the Bass's formula and the Chouinard's formula in commutative algebra for modules of finite quasi-injective dimension (see [7,12]).

We start by recalling some elementary properties of quasi-projective dimension of objects in abelian categories established in [8]. *Throughout this section, let $\mathcal{A}$ be an arbitrary abelian category with enough projective objects.* For each object $M \in \mathcal{A}$, we denote by $\mathrm{pd}_{\mathcal{A}} M$, $\mathrm{qpd}_{\mathcal{A}} M$ and $\Omega^n(M)$ its projective dimension, quasi-projective dimension and n-th syzygy for $n \geqslant 0$ in $\mathcal{A}$, respectively. For the definitions of quasi-projective resolution and quasi-projective dimension, we refer to Definition 2.1.

Theorem 1.1. [8, Propositions 3.3 and 3.6(2)] *The following statements hold for an object $M \in \mathcal{A}$.*

(1) $\mathrm{qpd}_{\mathcal{A}}(M \oplus N) \leqslant \sup\{\mathrm{qpd}_{\mathcal{A}} M, \mathrm{qpd}_{\mathcal{A}} N\}$ *for any object $N \in \mathcal{A}$. In particular, if N is projective, then* $\mathrm{qpd}_{\mathcal{A}}(M \oplus N) \leqslant \mathrm{qpd}_{\mathcal{A}} M$.

(2) $\mathrm{qpd}_{\mathcal{A}} \Omega(M) \leqslant \mathrm{qpd}_{\mathcal{A}} M$.

(3) *If M is periodic, that is, there exists a positive integer r such that $\Omega^r(M) \simeq M$, then* $\mathrm{qpd}_{\mathcal{A}} M = 0$.

From Theorem 1.1, we can see some obvious differences between projective dimension and quasi-projective dimension. For example, the inequalities in Theorem 1.1(1) are equalities if quasi-projective dimension is replaced by projective dimension, while a periodic, non-projective object has infinite projective dimension.

The paper is devoted to providing more homological properties for quasi-projective dimension of objects in abelian categories. Our main result reads as follows.

Theorem 1.2. *Let $\mathcal{A}$ be an abelian category with enough projective objects. The following statements hold for an object $M \in \mathcal{A}$.*

(1) *If* $\mathrm{pd}_{\mathcal{A}} M < \infty$, *then* $\mathrm{qpd}_{\mathcal{A}} M = \mathrm{pd}_{\mathcal{A}} M$.

(2) *If* $\mathrm{qpd}_{\mathcal{A}} M < \infty$ *and* $\mathrm{Ext}^n_{\mathcal{A}}(M, M) = 0$ *for all* $n \geqslant 2$, *then* $\mathrm{pd}_{\mathcal{A}} M < \infty$.

(3) *If $0 \to N \to E \to M \to 0$ is an exact sequence in $\mathcal{A}$ such that E is projective-injective, then*

$$\mathrm{qpd}_{\mathcal{A}} M \leqslant \mathrm{qpd}_{\mathcal{A}} N + 1.$$

(4) *Let $P_\bullet$ be a deleted projective resolution of M in $\mathcal{A}$. Suppose that there is an integer $n \geqslant 2$ and a chain map $g_\bullet : P_\bullet \to P_\bullet[n]$ of complexes over $\mathcal{A}$:*

$$\begin{array}{ccccccccccccc}
\cdots & \longrightarrow & P_{n+1} & \longrightarrow & P_n & \longrightarrow & P_{n-1} & \longrightarrow & \cdots & \longrightarrow & P_0 & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle g_{n+1}} & & \downarrow{\scriptstyle g_n} & & \downarrow{\scriptstyle 0} & & & & \downarrow{\scriptstyle 0} & & \\
\cdots & \longrightarrow & P_1 & \longrightarrow & P_0 & \longrightarrow & 0 & \longrightarrow & \cdots & \longrightarrow & 0 & \longrightarrow & 0
\end{array}$$

inducing an isomorphism $\Omega^{n+m}(M) \to \Omega^m(M)$ *for some integer* $m \geqslant 0$*. Then* $\operatorname{qpd}_{\mathcal{A}} M \leqslant m$*.*

Note that Theorem 1.2(1) and Theorem 1.2(2) have been shown for quasi-projective dimension of finitely generated modules over commutative Noetherian rings by using some techniques from commutative algebra; see [8, Corollary 4.10 and Theorem 6.20]. Theorem 1.2(3)(4) are completely new (even in the case of commutative rings), and provide two effective methods for bounding quasi-projective dimension (see Example 3.10). However, Theorem 1.2(3) does not hold when the object E is projective but not injective; see Example 3.7 for a counterexample.

An interesting application of Theorem 1.2(2) is the validity of *the Auslander–Reiten conjecture* for modules of finite quasi-projective dimension over left Noetherian rings. Recall that the conjecture says that if a finitely generated R-module M over a left Noetherian ring R satisfies $\operatorname{Ext}^n_R(M, M \oplus R) = 0$ for all $n \geqslant 1$, then it is projective.

We formulate the following corollary for left coherent rings which are a generalization of left Noetherian rings.

Corollary 1.3. *Let R be a left coherent ring. Suppose that M is a finitely presented left R-module with finite quasi-projective dimension. If* $\operatorname{Ext}^n_R(M, M \oplus R) = 0$ *for all* $n \geqslant 1$*, then M is projective.*

Analogous to global dimension, we introduce the notion of quasi-global dimension for left Noetherian rings and establish some basic properties of this new dimension in Proposition 4.2. In particular, we obtain two applications of this dimension to homological conjectures in the representation theory of Artin algebras:

(*a*) *The finitistic dimension conjecture holds for every Artin algebra with finite quasi-global dimension* (see Proposition 4.2(1));

(*b*) *Tachikawa's second conjecture holds for any self-injective Artin algebra with finite quasi-global dimension, or equivalently, with quasi-global dimension* 0 (see Corollary 4.3 and Remark 4.4).

Note that if two finite-dimensional self-injective algebras over a field are stably equivalent of Morita type or derived equivalent, then they have the same quasi-global dimension (see Corollary 4.6). For other new advances on Tachikawa's second conjecture, we also refer the reader to [4,6].

Further, using Theorem 1.2, we can also determine the quasi-global dimension of a class of finite-dimensional Nakayama algebras (see Theorem 4.8 for more details). Consequently, those algebras always have finite quasi-global dimension, although they may have infinite global dimension in some cases.

The paper is organized as follows. In Section 2, we recall some definitions and basic facts on quasi-projective dimension of objects in abelian categories. In Section 3, we show Theorem 1.2. In Section 4, we discuss quasi-global dimension for left Noetherian rings and calculate this dimension for a class of Nakayama algebras.

2. Preliminaries

In this section, we introduce some standard notation and recall the definitions of quasi-projective resolution and quasi-projective dimension of objects in abelian categories from [8].

All rings considered in this paper are assumed to be associative and with identity and all modules are unitary left modules.

Let R be a ring. We denote by R-Mod the category of all R-modules and by R-mod the category of all finitely presented R-modules. For an R-module M, let $\mathrm{add}(M)$ (respectively, $\mathrm{Add}(M)$) be the full subcategory of R-Mod consisting of all direct summands of finite (respectively, arbitrary) direct sums of copies of M. In many circumstances, we write R-proj and R-Proj for $\mathrm{add}({}_RR)$ and $\mathrm{Add}({}_RR)$, respectively. When R is *left coherent*, that is, every finitely generated left ideal of R is finitely presented, the category R-mod is an abelian category and has R-proj as its full subcategory consisting of all projective objects.

Let $\mathscr{A}$ be an additive category. By a *complex* $X_\bullet := (X_i, d_i^X)_{i\in\mathbb{Z}}$ over $\mathscr{A}$, we mean a sequence of morphisms d_i^X between objects X_i in $\mathscr{A}$:

$$\cdots \xrightarrow{d_{i+2}^X} X_{i+1} \xrightarrow{d_{i+1}^X} X_i \xrightarrow{d_i^X} X_{i-1} \xrightarrow{d_{i-1}^X} \cdots,$$

such that $d_i^X d_{i+1}^X = 0$ for $i \in \mathbb{Z}$. For simplicity, we sometimes write $(X_i)_{i\in\mathbb{Z}}$ for $X_\bullet$ without mentioning d_i^X. For a fixed integer n, we denote by $X_\bullet[n]$ the complex obtained from $X_\bullet$ by shifting n degrees, that is, $(X_\bullet[n])_i = X_{i-n}$ and $d_i^{X[n]} = (-1)^n d_{i-n}^X$. Let $\mathscr{C}(\mathscr{A})$ be the category of all complexes over $\mathscr{A}$ with chain maps, and let $\mathscr{K}(\mathscr{A})$ be the homotopy category of $\mathscr{C}(\mathscr{A})$. For a chain map $f_\bullet : X_\bullet \to Y_\bullet$ in $\mathscr{C}(\mathscr{A})$, we denote by $\mathrm{Con}(f_\bullet)$ the mapping cone of $f_\bullet$. When $\mathscr{A}$ is an abelian category, we also define the i-th cycle, boundary and homology of $X_\bullet$ for each i as follows:

$$Z_i(X_\bullet) := \mathrm{Ker}(d_i^X), \quad B_i(X_\bullet) := \mathrm{Im}(d_{i+1}^X) \quad \text{and} \quad H_i(X_\bullet) := Z_i(X_\bullet)/B_i(X_\bullet).$$

Moreover, the *supremum*, *infimum*, *homological supremum* and *homological infimum* of $X_\bullet$ are defined by

$$\sup(X_\bullet) := \sup\{i \in \mathbb{Z} \mid X_i \neq 0\}, \quad \inf(X_\bullet) := \inf\{i \in \mathbb{Z} \mid X_i \neq 0\},$$

$$\mathrm{hsup}(X_\bullet) := \sup\{i \in \mathbb{Z} \mid H_i(X_\bullet) \neq 0\}, \quad \mathrm{hinf}(X_\bullet) := \inf\{i \in \mathbb{Z} \mid H_i(X_\bullet) \neq 0\}.$$

Clearly $\inf(X_\bullet) \leqslant \mathrm{hinf}(X_\bullet) \leqslant \mathrm{hsup}(X_\bullet) \leqslant \sup(X_\bullet)$. We say that $X_\bullet$ is *bounded* (respectively, *bounded below*) if $\sup(X_\bullet) < \infty$ and $\inf(X_\bullet) > -\infty$ (respectively, $\inf(X_\bullet) > -\infty$). Let $\mathscr{C}^-(\mathcal{A})$ and $\mathscr{C}^{\mathrm{b}}(\mathcal{A})$ be the categories of all bounded below and bounded complexes over $\mathcal{A}$, respectively. Their homotopy categories are denoted by $\mathscr{K}^-(\mathcal{A})$ and $\mathscr{K}^{\mathrm{b}}(\mathcal{A})$, respectively.

When $\mathcal{A}$ is an abelian category, the bounded below derived category of $\mathcal{A}$ is denoted by $\mathscr{D}^-(\mathcal{A})$, which is the localization of $\mathscr{K}^-(\mathcal{A})$ at all quasi-isomorphisms. Moreover, the *extension* of two full subcategories $\mathcal{X}$ and $\mathcal{Y}$ in $\mathscr{D}^-(\mathcal{A})$, denoted by $\mathcal{X} * \mathcal{Y}$, is defined to be the full subcategory of $\mathscr{D}^-(\mathcal{A})$ consisting of all objects Z such that there exists a distinguished triangle $X \to Z \to Y \to X[1]$ in $\mathscr{D}^-(\mathcal{A})$ with $X \in \mathcal{X}$ and $Y \in \mathcal{Y}$. Normally, we identify $\mathcal{A}$ with the full subcategory of $\mathscr{D}^-(\mathcal{A})$ consisting of all stalk complexes concentrated in degree 0.

From now on, *let $\mathcal{A}$ be an abelian category with enough projective objects.*

Let M be an object of $\mathcal{A}$. We denote by $\mathrm{pd}_{\mathcal{A}} M$ the projective dimension of M in $\mathcal{A}$. Let

$$P_\bullet : \quad \cdots \longrightarrow P_{i+1} \longrightarrow P_i \longrightarrow P_{i-1} \longrightarrow \cdots \longrightarrow P_1 \longrightarrow P_0 \longrightarrow 0$$

be a deleted projective resolution of M in $\mathcal{A}$. For each integer $i \geqslant 1$, the i-th syzygy of M (with respect to $P_\bullet$) is defined to be $\Omega^i(M) := B_{i-1}(P_\bullet)$. As usual, we write $\Omega(M)$ for $\Omega^1(M)$, and understand that $\Omega^0(M) = M$. The object M is said to be *periodic* if there exists an integer $r \geqslant 1$ such that $\Omega^r(M) \simeq M$. When $\mathcal{A} = R$-Mod, we simply write $\mathrm{pd}_R M$ for $\mathrm{pd}_{\mathcal{A}} M$.

The following definition is taken from [8, Definition 3.1].

Definition 2.1. (1) A complex $P_\bullet \in \mathscr{C}^-(\mathcal{A})$ is called a *quasi-projective resolution* of M if the following two conditions hold:

(a) all P_i are projective for $i \in \mathbb{Z}$;

(b) for all integers $j \geqslant \inf(P_\bullet)$, there are integers $n_j \geqslant 0$, not all zero, such that $H_j(P_\bullet) \simeq M^{n_j}$. If, in addition, $\sup(P_\bullet) < \infty$ (that is, $P_\bullet \in \mathscr{C}^{\mathrm{b}}(\mathcal{A})$), then $P_\bullet$ is called a *finite quasi-projective resolution* of M.

(2) The *quasi-projective dimension* of a nonzero object M in $\mathcal{A}$ is defined to be

$$\mathrm{qpd}_{\mathcal{A}} M := \inf\{\sup(P_\bullet) - \mathrm{hsup}(P_\bullet) \mid P_\bullet \text{ is a finite quasi-projective resolution of } M\}.$$

We understand that the quasi-projective dimension of the zero object is 0 (compared with [8, Definition 3.1(2)]).

By Definition 2.1, $\mathrm{qpd}_{\mathcal{A}} M = \infty$ if and only if M does not have a finite quasi-projective resolution. Since each deleted projective resolution of M is automatically a quasi-projective resolution, it is clear that $\mathrm{qpd}_{\mathcal{A}} M \leqslant \mathrm{pd}_{\mathcal{A}} M$. The inequality can be strict, for example, if M is periodic but non-projective (see Theorem 1.1(3)). Moreover, by shifting quasi-projective resolutions, we see that $\mathrm{qpd}_{\mathcal{A}}(M)$ is the infimum of the projective dimensions $\mathrm{pd}_{\mathcal{A}}(\mathrm{Coker}(d_1^P)$, where $P_\bullet := (P_i, d_i^P)_{i\in\mathbb{Z}}$ runs through all finite quasi-projective resolutions of M with $\mathrm{hsup}\, P_\bullet = 0$. In this paper, for a ring R and any R-module M, we set $\mathrm{qpd}_R\, M := \mathrm{qpd}_{R\text{-Mod}}\, M$.

The following result reveals a close relationship between objects with finite quasi-projective dimension and objects with finite projective dimension.

Proposition 2.2. *Let $\mathcal{P}$ be the full subcategory of $\mathcal{A}$ consisting of (not necessarily all) projective objects and being closed under direct summands and finite direct sums. Let M be an object of $\mathcal{A}$ such that $\mathrm{qpd}_{\mathcal{A}}(M) < \infty$. Suppose that M admits a projective resolution of the form*

$$Q_\bullet : \quad \cdots \longrightarrow Q_2 \longrightarrow Q_1 \longrightarrow Q_0 \longrightarrow M \longrightarrow 0$$

with all $Q_i \in \mathcal{P}$ for $i \geqslant 0$. Then there exists a bounded complex

$$P_\bullet : \quad 0 \longrightarrow P_r \xrightarrow{d_r} \cdots \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{d_0} P_{-1} \longrightarrow \cdots \xrightarrow{d_{-s+1}} P_{-s} \longrightarrow 0$$

in $\mathscr{C}^{\mathrm{b}}(\mathcal{P})$ with $s \geqslant 0$ and $\mathrm{hsup}\, P_\bullet = 0$ satisfying the following:

(a) The complex $P_\bullet$ is a quasi-projective resolution of M and $\mathrm{qpd}_{\mathcal{A}}\, M = \mathrm{pd}_{\mathcal{A}}\, N$, where $N := \mathrm{Coker}(d_1)$.

(b) For a right exact, covariant functor $F : \mathcal{A} \to \mathcal{B}$ between abelian categories, let L^iF denote the i-th left-derived functor of F and let $d := \sup\{i \geqslant 0 \mid L^iF(M) \neq 0\}$.

If $0 \leqslant d < \infty$, then $d = \sup\{i \geqslant 0 \mid L^iF(N) \neq 0\}$. Furthermore, if $0 < d < \infty$, then $L^dF\big(H_0(P_\bullet)\big) \simeq L^dF(N)$.

Proof. Let $\mathcal{X}$ denote the full subcategory of $\mathcal{A}$ consisting of all projective objects. Then $\mathcal{P} \subseteq \mathcal{X}$. Let $r := \mathrm{qpd}_{\mathcal{A}} M < \infty$. Then M has a finite quasi-projective resolution $S_\bullet := (S_i, d_i^S)_{i\in\mathbb{Z}} \in \mathscr{C}^{\mathrm{b}}(\mathcal{X})$ in $\mathcal{A}$ with $\mathrm{hsup}(S_\bullet) = 0$ and $r = \mathrm{pd}_{\mathcal{A}}\big(\mathrm{Coker}(d_1^S)\big)$. Set $s := -\mathrm{hinf}(S_\bullet)$. Then $0 \leqslant s < \infty$. By taking truncations of complexes over $\mathcal{A}$, we have

$$S_\bullet \in H_0(S_\bullet) * H_{-1}(S_\bullet)[-1] * \cdots * H_{-s}(S_\bullet)[-s] \subseteq \mathscr{D}^-(\mathcal{A}).$$

For each integer i, $H_i(S_\bullet) \simeq M^{n_i}$ for some integer $n_i \geqslant 0$. Since $\mathcal{P} \subseteq \mathcal{A}$ is closed under finite direct sums, we can replace M with its projective resolution $Q_\bullet \in \mathscr{C}^-(\mathcal{P})$ and construct a complex $T_\bullet \in \mathscr{C}^-(\mathcal{P})$ that is isomorphic in $\mathscr{D}^-(\mathcal{A})$ to $S_\bullet$. Consequently, $H^i(T_\bullet) \simeq H^i(S_\bullet)$ for each $i \in \mathbb{Z}$. This implies that $T_\bullet$ is also a quasi-projective resolution of M with $\mathrm{hsup}(T_\bullet) = 0$. Further, since $\mathscr{K}^-(\mathcal{X})$ is triangle equivalent to $\mathscr{D}^-(\mathcal{A})$ and $\mathscr{K}^-(\mathcal{P}) \subseteq \mathscr{K}^-(\mathcal{X})$, we have $T_\bullet \simeq S_\bullet$ in $\mathscr{K}^-(\mathcal{X})$. It follows that there exist contractible complexes $X_\bullet$ and $Y_\bullet$ in $\mathscr{C}^-(\mathcal{X})$ such that $T_\bullet \oplus X_\bullet \simeq S_\bullet \oplus Y_\bullet$ in $\mathscr{C}^-(\mathcal{X})$. In both sides of the isomorphism, we take the cokernels of complexes in degree 1 and obtain

$$\mathrm{Coker}(d_1^T) \oplus \mathrm{Coker}(d_1^X) \simeq \mathrm{Coker}(d_1^S) \oplus \mathrm{Coker}(d_1^Y).$$

Since $X_\bullet$ and $Y_\bullet$ are contractible, both $\mathrm{Coker}(d_1^X)$ and $\mathrm{Coker}(d_1^Y)$ belong to $\mathcal{X}$. Thus $\mathrm{pd}_{\mathcal{A}}(\mathrm{Coker}(d_1^T)) = \mathrm{pd}_{\mathcal{A}}(\mathrm{Coker}(d_1^S))$. Since $\mathrm{hsup}(T_\bullet) = 0$, the sequence

$$\cdots \longrightarrow T_r \longrightarrow T_{r-1} \longrightarrow \cdots \longrightarrow T_1 \longrightarrow T_0 \longrightarrow \mathrm{Coker}(d_1^T) \longrightarrow 0,$$

truncated from the complex $T_\bullet$, is a projective resolution of $\mathrm{Coker}(d_1^T)$. It follows that $\Omega^r(\mathrm{Coker}(d_1^T))$ is projective, and thus isomorphic to a direct summand of T_r. Since $\mathcal{P}$ is closed under direct summands in $\mathcal{A}$ and contains T_r, we have $\Omega^r(\mathrm{Coker}(d_1^T)) \in \mathcal{P}$. Now, let $P_\bullet$ be the truncation of $T_\bullet$ in degree r:

$$\begin{aligned} P_\bullet :\ 0 \longrightarrow \Omega^r(\mathrm{Coker}(d_1^T)) \longrightarrow T_{r-1} \to T_{r-2} \longrightarrow \cdots \longrightarrow T_1 \longrightarrow T_0 \longrightarrow \cdots \\ \longrightarrow T_{-s} \longrightarrow 0. \end{aligned}$$

Then $P_\bullet \in \mathscr{C}^{\mathrm{b}}(\mathcal{P})$ is homotopy equivalent to $T_\bullet$ and satisfies (a).

To show (b), suppose $0 \leqslant d < \infty$. If $s = 0$, then $N \simeq H_0(P_\bullet) \simeq M^{n_0}$ and thus (b) holds trivially. Suppose $s > 0$. Observe that $H_0(P_\bullet) \simeq H_0(S_\bullet) \simeq M^{n_0} \neq 0$. Applying the functor $L^{d+j}F$ for $j \geqslant 0$ to the exact sequence $0 \to M^{n_0} \to N \to \mathrm{Im}(d_0) \to 0$ in $\mathcal{A}$ yields an exact sequence in $\mathcal{B}$:

$$L^{d+j+1}F\big(\mathrm{Im}(d_0)\big) \longrightarrow L^{d+j}F(M^{n_0}) \longrightarrow L^{d+j}F(N) \longrightarrow L^{d+j}F(\mathrm{Im}\, d_0).$$

Consequently, if $L^{d+j}F\big(\mathrm{Im}(d_0)\big) = 0$ for all $j > 0$, then

$$0 = \big(L^{d+j}F(M)\big)^{n_0} \simeq L^{d+j}F(M^{n_0}) \simeq L^{d+j}F(N) \quad \text{and} \quad 0 \neq L^dF(M^{n_0}) \hookrightarrow L^dF(N),$$

and therefore $d = \sup\{i \geqslant 0 \mid L^iF(N) \neq 0\}$.

It remains to show $L^{d+j}F\big(\mathrm{Im}(d_0)\big) = 0$ for all $j > 0$.

Now, we fix $j > 0$. Since $L^{d+k}F(M) = 0$ for all $k > 0$ and $H_{i-1}(P_\bullet) \simeq M^{n_{i-1}}$ for $i \in \mathbb{Z}$, we apply the functor $L^{d+j}F$ to the exact sequence $0 \to \mathrm{Im}(d_i) \to \mathrm{Ker}(d_{i-1}) \to H_{i-1}(P_\bullet) \to 0$ and obtain

$$L^{d+j}F(\mathrm{Im}(d_i)) \simeq L^{d+j}F(\mathrm{Ker}(d_{i-1})) \ \text{ and } \ L^dF(\mathrm{Im}(d_i)) \hookrightarrow L^dF(\mathrm{Ker}(d_{i-1})).$$

Note that $\mathrm{Ker}(d_{i-1}) = \Omega(\mathrm{Im}(d_{i-1}))$ due to $P_{i-1} \in \mathcal{P}$. It follows that $L^mF(\mathrm{Ker}(d_{i-1})) \simeq L^{m+1}F(\mathrm{Im}(d_{i-1}))$ for all $m > 0$. Thus $L^{d+j}F(\mathrm{Im}(d_i)) \simeq L^{d+j+1}F(\mathrm{Im}(d_{i-1}))$ and

$$L^dF(\mathrm{Im}(d_i)) \hookrightarrow L^dF(\mathrm{Ker}(d_{i-1})) \simeq L^{d+1}F(\mathrm{Im}(d_{i-1}))$$

for $d > 0$. This implies a series of isomorphisms:

$$\begin{aligned} L^{d+j}F(\mathrm{Im}(d_0)) \simeq L^{d+j+1}F(\mathrm{Im}(d_{-1})) \simeq L^{d+j+2}F(\mathrm{Im}(d_{-2})) \simeq \cdots \\ \simeq L^{d+j+s}F(\mathrm{Im}(d_{-s})) = 0, \end{aligned}$$

where the last equality follows from the vanishing of the map $d_{-s} : P_{-s} \to 0$.

Suppose $d > 0$. Then

$$L^d F(\mathrm{Im}(d_0)) \hookrightarrow L^{d+1} F(\mathrm{Im}(d_{-1})) \simeq L^{d+2} F(\mathrm{Im}(d_{-2})) \simeq \cdots \simeq L^{d+s} F(\mathrm{Im}(d_{-s})) = 0,$$

and therefore $L^d F(\mathrm{Im}(d_0)) = 0$. From the exact sequence

$$0 = L^{d+1} F(\mathrm{Im}(d_0)) \longrightarrow L^d F(M^{n_0}) \longrightarrow L^d F(N) \longrightarrow L^d F(\mathrm{Im}(d_0)) = 0,$$

we see that $L^d F(H_0(P_\bullet)) \simeq L^d F(M^{n_0}) \simeq L^d F(N)$. This shows (b).

Similarly, we can also show that if $d = -\infty$ (that is, $L^i F(M) = 0$ for all $i \geqslant 0$), then

$$\sup\{i \in \mathbb{N} \mid L^i F(N) \neq 0\} = \begin{cases} 0, & \text{if } F(N) \neq 0; \\ -\infty, & \text{if } F(N) = 0. \end{cases} \qquad \square$$

A direct consequence of Proposition 2.2 is the following result which generalizes [8, Proposition 3.4].

Corollary 2.3. *Let R be a left coherent ring and M a finitely presented R-module with finite quasi-projective dimension. Then:*

(1) *There exists a bounded complex $P_\bullet \in \mathscr{K}^{\mathrm{b}}(R\text{-proj})$ that is a quasi-projective resolution of M such that $\mathrm{qpd}_R(M) = \sup(P_\bullet) - \mathrm{hsup}(P_\bullet)$.*

(2) *There exists a finitely presented R-module N such that $\mathrm{pd}_R(N) = \mathrm{qpd}_R(M)$, and for which there is an injection $M \hookrightarrow N$ of R-modules.*

Proof. Since R is left coherent, R-mod is an abelian category and has R-proj as its full subcategory consisting of all projective objects. In Proposition 2.2, we take $\mathcal{A} = R$-mod and $\mathcal{P} = R$-proj. By Proposition 2.2(a), there exists a bounded complex $P_\bullet := (P_i, d_i^P)_{i \in \mathbb{Z}} \in \mathscr{C}^{\mathrm{b}}(R\text{-proj})$ satisfying (1) and $\mathrm{hsup}(P_\bullet) = 0$. Let $N := \mathrm{Coker}(d_1^P)$. Then $N \in R$-mod and $\mathrm{pd}_R(N) = \mathrm{qpd}_R(M)$. Since there is an exact sequence $0 \to H_0(P_\bullet) \to N \to \mathrm{Im}(d_0^P) \to 0$ of R-modules with $H_0(P_\bullet) \simeq M^{n_0}$ for some integer $n_0 > 0$, we obtain an injection $M \hookrightarrow N$ in R-mod. $\square$

Corollary 2.4. *Let $\mathcal{A}$ be a Frobenius abelian category. Then $\mathrm{qpd}_{\mathcal{A}}(M) = 0$ or ∞ for any $M \in \mathcal{A}$.*

Proof. Since $\mathcal{A}$ is a Frobenius category, projective objects and injective objects of $\mathcal{A}$ coincide. This implies that objects of $\mathcal{A}$ with finite projective dimension are exactly projective objects. By Proposition 2.2(a), if $\mathrm{qpd}_{\mathcal{A}}(M) < \infty$, then $\mathrm{qpd}_{\mathcal{A}}(M) = 0$. $\square$

Finally, we apply Proposition 2.2 to give a new proof of [9, Theorem 4.2] which is a generalization of the classical depth formula (see [2, Theorem 1.2]).

The following definition is standard in commutative algebra.

Definition 2.5. Let $(R, \mathfrak{m})$ be a commutative Noetherian local ring with the maximal ideal $\mathfrak{m}$. The *depth* of a finitely generated R-module M is defined as

$$\mathrm{depth}_R(M) \coloneqq \min\{i \geqslant 0 \mid \mathrm{Ext}^i_R(R/\mathfrak{m}, M) \neq 0\}.$$

Theorem 2.6. [9, Theorem 4.2] *Let R be a commutative Noetherian local ring, and let M and N be non-zero finitely generated R-modules. Define $q := \sup\{i \geqslant 0 \mid \mathrm{Tor}^R_i(M, N) \neq 0\}$. Suppose that $q < \infty$ and $\mathrm{qpd}_R(M) < \infty$. If $q = 0$ or $\mathrm{depth}(\mathrm{Tor}^R_q(M, N)) \leqslant 1$, then*

$$\mathrm{depth}(N) = \mathrm{depth}(\mathrm{Tor}^R_q(M, N)) + \mathrm{qpd}_R(M) - q.$$

Proof. The case $q = 0$ in Theorem 2.6 (that is, $\mathrm{Tor}^R_i(M, N) = 0$ for all $i > 0$) is a direct consequence of [8, Theorems 4.4 and 4.11].

For the case $q > 0$, we provide a new proof that is different from the one given in [9, Theorem 4.2].

Let $F := - \otimes_R N : R\text{-Mod} \to R\text{-Mod}$, a covariant and right exact functor. By definition, for each $i \in \mathbb{N}$, the i-th left-derived functor L^iF of F is exactly the functor $\mathrm{Tor}^R_i(-, N)$. This implies

$$q = \sup\{i \geqslant 0 \mid L^i(F)(M) \neq 0\} < \infty.$$

Since $\mathrm{qpd}_R(M) < \infty$, it follows from Proposition 2.2(b) that there exists a finite quasi-projective resolution $(P_\bullet, d_i)_{i\in\mathbb{Z}}$ of ${}_RM$ with $\mathrm{hsup}\, P_\bullet = 0$ and $C := \mathrm{Coker}(d_1)$ satisfying

$$q = \sup\{i \geqslant 0 \mid \mathrm{Tor}^R_i(C, N) \neq 0\}, \quad \mathrm{pd}_R(C) = \mathrm{qpd}_R(M) < \infty \quad \text{and}$$
$$\mathrm{Tor}^R_q(C, N) \simeq \mathrm{Tor}^R_q(H_0(P_\bullet), N).$$

Note that $H_0(P_\bullet) \simeq M^{n_0}$ for some $n_0 > 0$. This implies $\mathrm{depth}(\mathrm{Tor}^R_q(C, N)) = \mathrm{depth}(\mathrm{Tor}^R_q(M, N)) \leqslant 1$. Now, we apply *Auslander's depth formula* (see [2, Theorem 1.2]) directly to the pair (C, N) of R-modules, and obtain the formula

$$\mathrm{depth}(N) = \mathrm{depth}(\mathrm{Tor}^R_q(C, N)) + \mathrm{pd}_R(C) - q.$$

Thus $\mathrm{depth}(N) = \mathrm{depth}(\mathrm{Tor}^R_q(M, N)) + \mathrm{qpd}_R(M) - q.$ $\square$

3. New properties of quasi-projective dimension

In this section, we develop some general methods to determine the quasi-projective dimension of objects in abelian categories. In particular, we give a proof of Theorem 1.2.

Throughout this section, $\mathcal{A}$ stands for an abelian category with enough projective objects. Denote by $\mathcal{P}$ the full subcategory of $\mathcal{A}$ consisting of *all* projective objects.

Our first result generalizes [8, Corollary 4.10] that is focused on finitely generated modules over commutative Noetherian rings and is a corollary of the *Auslander–Buchsbaum formula* for modules of finite quasi-projective dimension established in [8, Theorem 4.4].

Proposition 3.1. *Let M be an object of $\mathcal{A}$ with $\operatorname{pd}_{\mathcal{A}} M < \infty$. Then $\operatorname{qpd}_{\mathcal{A}} M = \operatorname{pd}_{\mathcal{A}} M$.*

Proof. Note that $\operatorname{qpd}_{\mathcal{A}} M \leqslant \operatorname{pd}_{\mathcal{A}} M < \infty$. It suffices to show $\operatorname{pd}_{\mathcal{A}} M \leqslant \operatorname{qpd}_{\mathcal{A}} M$. Let N be the object given in Proposition 2.2(a). Then $\operatorname{qpd}_{\mathcal{A}} M = \operatorname{pd}_{\mathcal{A}} N$. We claim $\operatorname{pd}_{\mathcal{A}} M \leqslant \operatorname{pd}_{\mathcal{A}} N$.

In fact, for each object $X \in \mathcal{A}$, the contravariant functor $F_X := \operatorname{Hom}_{\mathcal{A}}(-, X) : \mathcal{A} \to \mathbb{Z}\text{-Mod}$ can be regarded as a right exact, covariant functor $\mathcal{A} \to (\mathbb{Z}\text{-Mod})^{\text{op}}$, the opposite category of $\mathbb{Z}$-Mod. Then, for each $i \geqslant 0$, the i-th left-derived functor $L^i F_X$ of F_X is exactly the i-th extension functor $\operatorname{Ext}^i_{\mathcal{A}}(-, X)$. Let $d_X := \sup\{i \geqslant 0 \mid \operatorname{Ext}^i_{\mathcal{A}}(M, X) \neq 0\}$. By Proposition 2.2($b$), if $0 \leqslant d_X < \infty$, then $d_X = \sup\{i \geqslant 0 \mid \operatorname{Ext}^i_{\mathcal{A}}(N, X) \neq 0\}$. Let $n := \operatorname{pd}_{\mathcal{A}} M < \infty$. Then $\operatorname{Ext}^i_{\mathcal{A}}(M, -) = 0$ for $i > n$ and there exists an object $X \in \mathcal{A}$ such that $\operatorname{Ext}^n_{\mathcal{A}}(M, X) \neq 0$. This implies $d_X = n$. Thus $\operatorname{Ext}^n_{\mathcal{A}}(N, X) \neq 0$ which forces $\operatorname{pd}_{\mathcal{A}} N \geqslant n$. □

Lemma 3.2. *Let M be an object of $\mathcal{A}$ with $\operatorname{Ext}^n_{\mathcal{A}}(M, M) = 0$ for $2 \leqslant n \leqslant k+1$ with $k \in \mathbb{N}$. Suppose that $X_\bullet \in \mathscr{D}^{\mathrm{b}}(\mathcal{A})$ satisfies that $H_i(X_\bullet) \in \operatorname{add}(M)$ for $0 \leqslant i \leqslant k$ and $H_i(X_\bullet) = 0$ for $i > k$ or $i < 0$. Then $X_\bullet \simeq \bigoplus_{i=0}^{k} H_i(X_\bullet)[i]$ in $\mathscr{D}^{\mathrm{b}}(\mathcal{A})$.*

Proof. We take induction on k to show Lemma 3.2.

Lemma 3.2 holds trivially for $k = 0$. Let $k > 0$. Taking the canonical truncation of $X_\bullet$ at degree k yields a distinguished triangle in $\mathscr{D}^{\mathrm{b}}(\mathcal{A})$:

$$\tau_{\geqslant k} X_\bullet \longrightarrow X_\bullet \longrightarrow \tau_{\leqslant k-1} X_\bullet \xrightarrow{f} (\tau_{\geqslant k} X_\bullet)[1]$$

where $\tau_{\geqslant k} X_\bullet \simeq H_k(X_\bullet)[k] \in \operatorname{add}(M[k])$ and $H_i(\tau_{\leqslant k-1} X_\bullet) \simeq H_i(X_\bullet)$ for $i \leqslant k-1$ but $H_i(\tau_{\leqslant k-1} X_\bullet) = 0$ for $i \geqslant k$. In particular, $H_i(\tau_{\leqslant k-1} X_\bullet) \in \operatorname{add}(M)$ for $0 \leqslant i \leqslant k-1$. Now, induction on $k-1$ implies that

$$\tau_{\leqslant k-1} X_\bullet \simeq \bigoplus_{i=0}^{k-1} H_i(\tau_{\leqslant k-1} X_\bullet)[i] \simeq \bigoplus_{i=0}^{k-1} H_i(X_\bullet)[i] \in \operatorname{add}(\bigoplus_{i=0}^{k-1} M[i]).$$

Since $\operatorname{Ext}^n_{\mathcal{A}}(M, M) = 0$ for $2 \leqslant n \leqslant k+1$, we have $\operatorname{Hom}_{\mathscr{D}^{\mathrm{b}}(\mathcal{A})}(\bigoplus_{i=0}^{k-1} M[i], M[k+1]) = 0$. This forces $\operatorname{Hom}_{\mathscr{D}^{\mathrm{b}}(\mathcal{A})}(\tau_{\leqslant k-1} X_\bullet, (\tau_{\geqslant k} X_\bullet)[1]) = 0$. Thus $f = 0$ and $X_\bullet \simeq \tau_{\geqslant k} X_\bullet \oplus \tau_{\leqslant k-1} X_\bullet \simeq \bigoplus_{i=0}^{k} H_i(X_\bullet)[i]$ in $\mathscr{D}^{\mathrm{b}}(\mathcal{A})$. □

Theorem 3.3. *Suppose that $M \in \mathcal{A}$ has a finite quasi-projective resolution $P_\bullet$. If $\operatorname{Ext}^n_{\mathcal{A}}(M, M) = 0$ for $2 \leqslant n \leqslant \operatorname{hsup}(P_\bullet) - \operatorname{hinf}(P_\bullet) + 1$, then $\operatorname{pd}_{\mathcal{A}}(M) < \infty$.*

Proof. Let $t := \operatorname{hsup}(P_\bullet)$ and $s := \operatorname{hinf}(P_\bullet)$. Then t and s are finite with $s \leqslant t$. Since $P_\bullet$ is a finite quasi-projective resolution of M, we see that $P_\bullet \in \mathscr{C}^{\mathrm{b}}(\mathcal{P})$, $H_i(P_\bullet) \simeq M^{a_i}$ for $s \leqslant i \leqslant t$ and $H_i(P_\bullet) = 0$ for $i > t$ or $i < s$, where a_i are nonnegative integers and not all zero. Suppose that $\operatorname{Hom}_{\mathscr{D}^{-}(\mathcal{A})}(M, M[n]) = 0$ for $2 \leqslant n \leqslant t - s + 1$. By Lemma 3.2,

$P_\bullet \simeq \bigoplus_{i=s}^t H_i(P_\bullet)[i] \simeq \bigoplus_{i=s}^t M^{a_i}[i]$ in $\mathscr{D}^-(\mathcal{A})$. Now, let $Q_\bullet$ be a deleted projective resolution of M. Then there is an isomorphism $P_\bullet \simeq \bigoplus_{i=s}^t Q_\bullet^{a_i}[i]$ in $\mathscr{D}^-(\mathcal{A})$, where both sides are bounded below complexes of projective objects. Since the localization functor $\mathscr{K}^-(\mathcal{P}) \to \mathscr{D}^-(\mathcal{A})$ is a triangle equivalence, it follows that $P_\bullet \simeq \bigoplus_{i=s}^t Q_\bullet^{a_i}[i]$ in $\mathscr{K}^-(\mathcal{P})$. Note that $\mathscr{K}^{\mathrm{b}}(\mathcal{P})$ is closed under direct summands in $\mathscr{K}^-(\mathcal{P})$. Since $P_\bullet \in \mathscr{K}^{\mathrm{b}}(\mathcal{P})$ and at least one a_i is not zero, we have $Q_\bullet \in \mathscr{K}^{\mathrm{b}}(\mathcal{P})$. This implies $\mathrm{pd}_{\mathcal{A}}(M) < \infty$. □

Theorem 3.3 implies the following result.

Corollary 3.4. *Let $M \in \mathcal{A}$. If $\mathrm{qpd}_{\mathcal{A}}(M) < \infty$ and $\mathrm{Ext}^n_{\mathcal{A}}(M, M) = 0$ for $n \geqslant 2$, then $\mathrm{pd}_{\mathcal{A}}(M) < \infty$.*

We present two direct corollaries of Corollary 3.4.

Corollary 3.5. *Let R be a left coherent ring and let M be a finitely presented R-module with finite quasi-projective dimension such that $\mathrm{Ext}^n_R(M, M \oplus R) = 0$ for all $n > 0$. Then M is projective.*

Proof. Let $\mathcal{A} := R$-mod. This is an abelian category with enough projective objects since R is left coherent. By Corollary 3.4, $\mathrm{pd}_R(M) < \infty$. Let $m := \mathrm{pd}_R(M)$. We claim $m = 0$.

In fact, since ${}_RM$ is finitely presented, it has a projective resolution $0 \to P_m \xrightarrow{f_m} P_{m-1} \to \cdots \to P_0 \to M \to 0$ with all $P_i \in R$-proj for $0 \leqslant i \leqslant m$. If $m > 0$, then the assumption $\mathrm{Ext}^m_R(M, R) = 0$ implies $\mathrm{Ext}^m_R(M, P_m) = 0$, and therefore f_m is a split injection. This leads to $\mathrm{pd}_R(M) < m$, a contradiction. Thus $m = 0$ and M is projective. □

Recall that a ring R is *quasi-Frobenius* if all projective left R-modules are injective, or equivalently, all injective left R-modules are projective. Note that if R is quasi-Frobenius, then it is left and right Noetherian and Artinian, and both R-Mod and R-mod are Frobenius abelian categories.

Corollary 3.6. *Let R be a quasi-Frobenius ring and let M be an R-module with finite quasi-projective dimension. If $\mathrm{Ext}^n_R(M, M) = 0$ for $n \geqslant 2$, then M is projective.*

Proof. By Corollary 3.4, $\mathrm{pd}_R(M) < \infty$. Since R is quasi-Frobenius, each R-module of finite projective dimension is projective. Thus M is projective. □

Corollary 3.4 provides a method for constructing modules with infinite quasi-projective dimension.

Example 3.7. Let A be an algebra over a field K presented by the quiver

$$1 \xrightarrow{\alpha} 2 \circlearrowright \beta$$

with relations: $\beta\alpha = 0 = \beta^2$, where the composition of arrows is always taken from right to left. Let S_1 and S_2 be simple A-modules corresponding to the vertices 1 and 2, respectively. It can be checked that $\Omega(S_1) = S_2$ and $\Omega(S_2) = S_2$. This implies $\mathrm{pd}_A(S_1) = \infty = \mathrm{pd}_A(S_2)$ and $\mathrm{Ext}^i_A(S_1, S_1) = 0$ for $i > 0$ (in fact, S_1 is injective). Thus $\mathrm{qpd}_A(S_1) = \infty$ by Corollary 3.4. Moreover, since S_2 is periodic, $\mathrm{qpd}_A(S_2) = 0$ by Theorem 1.1(3).

Note that there exists an exact sequence $0 \to S_2 \to P_1 \to S_1 \to 0$ of A-modules, where P_1 is the projective cover of S_1. Obviously, the inequality $\mathrm{qpd}_A(S_1) \leqslant \mathrm{qpd}_A(S_2)+1$ does not hold. But we always have $\mathrm{pd}_A(M) \leqslant \mathrm{pd}_A(\Omega(M)) + 1$ for any $M \in \mathscr{A}$. This also shows that quasi-projective dimension is different from projective dimension.

Next, we provide a sufficient condition for the inequality $\mathrm{qpd}_{\mathscr{A}}(M) \leqslant \mathrm{qpd}_{\mathscr{A}}(\Omega(M))+1$ to hold. Note that the inequality fails in Example 3.7 above.

Proposition 3.8. *Let $0 \to N \to E \to M \to 0$ be an exact sequence in $\mathscr{A}$ such that E is projective-injective. Then $\mathrm{qpd}_{\mathscr{A}}(M) \leqslant \mathrm{qpd}_{\mathscr{A}}(N) + 1$.*

Proof. The inequality in Proposition 3.8 automatically holds if $\mathrm{qpd}_{\mathscr{A}}(N) = \infty$. So, we assume $\mathrm{qpd}_{\mathscr{A}}(N) = m < \infty$. Then N admits a quasi-projective resolution of the form

$$P_\bullet: \quad 0 \longrightarrow P_m \xrightarrow{d_m} P_{m-1} \xrightarrow{d_{m-1}} \cdots \xrightarrow{d_1} P_0 \xrightarrow{d_0} \cdots \xrightarrow{d_{-r+1}} P_{-r} \xrightarrow{d_{-r}} 0,$$

where $P_m \neq 0$, $\mathrm{hsup}(P_\bullet) = 0$, $H_{-r}(P_\bullet) \neq 0$, and for each $i \in \mathbb{Z}$, there is an isomorphism $g_i : H_i(P_\bullet) \to N^{a_i}$ for some integers $a_i \in \mathbb{N}$ (not all zero). Let $0 \to N \xrightarrow{f} E \longrightarrow M \to 0$ be the exact sequence in Proposition 3.8. Since E is injective, the composition

$$\mathrm{Ker}(d_i) \twoheadrightarrow H_i(P_\bullet) \xrightarrow{g_i} N^{a_i} \xrightarrow{f^{a_i}} E^{a_i}$$

can be lifted to a morphism $h_i : P_i \longrightarrow E^{a_i}$ in $\mathscr{A}$. This is illustrated by the following commutative diagram

$$\begin{tikzcd}
 & P_{i+1} \arrow[dl, "d_{i+1}"'] \arrow[rrrr, two heads, "d_{i+1}"] & & & & P_i \arrow[d, "h_i"] \\
\mathrm{Im}(d_{i+1}) \arrow[r, hook] & \mathrm{Ker}(d_i) \arrow[r] \arrow[urrrr, hook] & H_i(P_\bullet) \arrow[r, "g_i"] & N^{a_i} \arrow[rr, "f^{a_i}"] & & E^{a_i}.
\end{tikzcd}$$

Note that $h_i d_{i+1} = 0$. Let

$$E_\bullet := 0 \longrightarrow E^{a_m} \xrightarrow{0} \cdots \xrightarrow{0} E^{a_0} \longrightarrow \cdots \xrightarrow{0} E^{a_{-r}} \longrightarrow 0$$

be a bounded complex over $\mathscr{A}$ with all differentials zero. Since E is projective, we have $E_\bullet \in \mathscr{C}^{\mathrm{b}}(\mathcal{P})$. Set $h_\bullet := (h_i)_{i\in\mathbb{Z}} : P_\bullet \longrightarrow E_\bullet$. Then $h_\bullet$ is a chain map in $\mathscr{C}^{\mathrm{b}}(\mathcal{P})$. Now, we extend the chain map to a distinguished triangle $P_\bullet \xrightarrow{h_\bullet} E_\bullet \longrightarrow \mathrm{Con}(h_\bullet) \longrightarrow P_\bullet[1]$ in

$\mathscr{K}^{\mathrm{b}}(\mathcal{P})$, where $\mathrm{Con}(h_\bullet)$ denotes the mapping cone of $h_\bullet$. By [13, Corollary 10.1.4], there is a long exact sequence

$$\cdots \longrightarrow H_i(P_\bullet) \xrightarrow{H_i(h_\bullet)} H_i(E_\bullet) \longrightarrow H_i(\mathrm{Con}(h_\bullet)) \longrightarrow H_{i-1}(P_\bullet) \xrightarrow{H_{i-1}(h_\bullet)} \cdots.$$

Observe that $H_i(h_\bullet) = f^{a_i} g_i$. Since f is a monomorphism and g_i is an isomorphism, we obtain a short exact sequence

$$0 \longrightarrow H_i(P_\bullet) \xrightarrow{H_i(h_\bullet)} H_i(E_\bullet) \longrightarrow H_i(\mathrm{Con}(h_\bullet)) \longrightarrow 0.$$

It follows from $\mathrm{Coker}(f) \simeq M$ that $H_i(\mathrm{Con}(h_\bullet)) \simeq M^{a_i}$. Thus $\mathrm{Con}(h_\bullet)$ is a finite quasi-projective resolution of M. By $\mathrm{hsup}(P_\bullet) = 0$, we have $a_i = 0$ for $i > 0$ and $a_0 \neq 0$. This forces $\mathrm{hsup}(\mathrm{Con}(h_\bullet)) = 0$. Note that $\sup(\mathrm{Con}(h_\bullet)) = m+1$ because the $(m+1)$-th term of the complex $\mathrm{Con}(h_\bullet)$ is exactly P_m. Therefore $\mathrm{qpd}_{\mathcal{A}}(M) \leqslant \sup(\mathrm{Con}(h_\bullet)) - \mathrm{hsup}(\mathrm{Con}(h_\bullet)) = m+1 = \mathrm{qpd}_{\mathcal{A}}(N) + 1$. $\square$

Corollary 3.9. *Let $\mathcal{A}$ be a Frobenius abelian category. For any $M \in \mathcal{A}$, $P \in \mathcal{P}$ and $i \in \mathbb{Z}$,*

$$\mathrm{qpd}_{\mathcal{A}}(M \oplus P) = \mathrm{qpd}_{\mathcal{A}}(M) = \mathrm{qpd}_{\mathcal{A}}(\Omega^i(M)).$$

Proof. By Theorem 1.1(1), $\mathrm{qpd}_{\mathcal{A}}(M \oplus P) \leqslant \mathrm{qpd}_{\mathcal{A}}(M)$. By Proposition 3.8, $\mathrm{qpd}_{\mathcal{A}}(M) \leqslant \mathrm{qpd}_{\mathcal{A}}(\Omega(M))+1$. Since $\Omega(M) = \Omega(M \oplus P)$, we have $\mathrm{qpd}_{\mathcal{A}}(\Omega(M)) = \mathrm{qpd}_{\mathcal{A}}(\Omega(M \oplus P)) \leqslant \mathrm{qpd}_{\mathcal{A}}(M \oplus P)$ by Theorem 1.1(2). Thus $\mathrm{qpd}_{\mathcal{A}}(M \oplus P) < \infty$ if and only if $\mathrm{qpd}_{\mathcal{A}}(M) < \infty$ if and only if $\mathrm{qpd}_{\mathcal{A}}(\Omega(M)) < \infty$. By Corollary 2.4, the objects $M \oplus P$, M and $\Omega(M)$ (and thus also $\Omega^i(M)$) have the same quasi-projective dimension. $\square$

Proof of Theorem 1.2. The statements (1)-(3) are Proposition 3.1, Corollary 3.4 and Proposition 3.8, respectively. It remains to show (4).

For each $s \geqslant 0$, we denote by $P_\bullet^{\leqslant s}$ the brutal truncation at degree s, that is,

$$P_i^{\leqslant s} = \begin{cases} P_i & \text{if } i \leqslant s; \\ 0 & \text{if } i > s. \end{cases}$$

The monomorphism $\Omega^{n+m}(M) \longrightarrow P_{n+m-1}$ induces a distinguished triangle in $\mathscr{D}^-(\mathcal{A})$:

$$\Omega^{n+m}(M)[n+m-1] \longrightarrow P_\bullet^{\leqslant n+m-1} \longrightarrow M \longrightarrow \Omega^{n+m}(M)[n+m],$$

where M is identified with the complex $0 \to \Omega^{n+m}(M) \to P_{n+m-1} \to P_{n+m-2} \to \cdots \to P_1 \to P_0 \to 0$, up to isomorphism. Similarly, there is a distinguished triangle in $\mathscr{D}^-(\mathcal{A})$:

$$\Omega^m(M)[m-1] \longrightarrow P_\bullet^{\leqslant m-1} \longrightarrow M \longrightarrow \Omega^m(M)[m].$$

Shifting this triangle by n steps yields a new distinguished triangle

$$\Omega^m(M)[n+m-1] \longrightarrow P_\bullet^{\leqslant m-1}[n] \longrightarrow M[n] \longrightarrow \Omega^m(M)[n+m].$$

Further, since the chain map $g_\bullet : P_\bullet \to P_\bullet[n]$ induces an isomorphism $\Omega^{n+m}(M) \to \Omega^m(M)$, we can construct the following commutative diagram in $\mathscr{D}^-(\mathcal{A})$:

$$\begin{tikzcd}
\Omega^{n+m}(M)[n+m-1] \arrow[r] \arrow[d, "\simeq"'] & P_\bullet^{\leqslant n+m-1} \arrow[r] \arrow[d, "g_\bullet^{\leqslant n+m-1}"] & M \arrow[r] \arrow[d] & \Omega^{n+m}(M)[n+m] \arrow[d, "\simeq"'] \\
\Omega^m(M)[n+m-1] \arrow[r] & P_\bullet^{\leqslant m-1}[n] \arrow[r] \arrow[d] & M[n] \arrow[r] \arrow[d] & \Omega^m(M)[n+m] \\
& \mathrm{Con}(g_\bullet^{\leqslant n+m-1}) \arrow[r, equal] \arrow[d] & \mathrm{Con}(g_\bullet^{\leqslant n+m-1}) \arrow[d] & \\
& P_\bullet^{\leqslant n+m-1}[1] \arrow[r] & M[1], &
\end{tikzcd}$$

where all columns and rows are distinguished triangles by the Octahedral axiom. Clearly $\mathrm{Con}(g_\bullet^{\leqslant n+m-1})$ lies in $\mathscr{C}^{\mathrm{b}}(\mathcal{P})$. By $n \geqslant 2$, we see that $H_i(\mathrm{Con}(g_\bullet^{\leqslant n+m-1})) \simeq M$ if $i = 1$ or n, and $H_i(\mathrm{Con}(g_\bullet^{\leqslant n+m-1})) = 0$ for other values of i. It follows that $\mathrm{Con}(g_\bullet^{\leqslant n+m-1})$ is a finite quasi-projective resolution of M. Thus $\mathrm{qpd}_{\mathcal{A}}(M) \leqslant \sup(\mathrm{Con}(g_\bullet^{\leqslant n+m-1})) - \mathrm{hsup}(\mathrm{Con}(g_\bullet^{\leqslant n+m-1})) = n+m-n = m$. $\square$

Example 3.10. Let A be an algebra over a field K presented by the quiver

$$\begin{tikzcd}
1 \arrow[r, "\alpha"] & 2 \arrow[d, "\beta"] \\
4 \arrow[u, "\delta"] & 3 \arrow[l, "\gamma"]
\end{tikzcd}$$

with relations: $\delta\gamma\beta\alpha = \beta\alpha\delta\gamma = 0$. Denote by J the Jacobson radical of A, and by e_i the primitive, idempotent element corresponding to the vertex i for $1 \leqslant i \leqslant 4$. Then there exists an obvious algebra automorphism $\Phi : A \to A$ sending $e_1 \mapsto e_3$, $e_2 \mapsto e_4$, $e_3 \mapsto e_1$ and $e_4 \mapsto e_2$.

We will show that

(1) $\mathrm{qpd}_A(Ae_1/Je_1) = \mathrm{qpd}_A(Ae_3/Je_3) = \mathrm{qpd}_A(Ae_2/J^2e_2) = \mathrm{qpd}_A(Ae_4/J^2e_4) = 2$.
(2) $\mathrm{qpd}_A(Ae_2/J^3e_2) = \mathrm{qpd}_A(Ae_4/J^3e_4) = 1$.

In fact, A is a Nakayama algebra, and thus is of finite representation type. By [1, Theorem V.3.5], every indecomposable A-module is isomorphic to P/J^tP for some indecomposable projective A-module P and for some integer t with $1 \leqslant t \leqslant 5$. Moreover, the indecomposable projective modules of A are determined by their composition sequences:

$$Ae_1 = \begin{matrix}1\\2\\3\\4\end{matrix}, \qquad Ae_2 = \begin{matrix}2\\3\\4\\1\\2\end{matrix}, \qquad Ae_3 = \begin{matrix}3\\4\\1\\2\end{matrix}, \qquad Ae_4 = \begin{matrix}4\\1\\2\\3\\4\end{matrix}.$$

Note that Ae_2 and Ae_4 are projective-injective, and that the indecomposable modules of projective dimension 1 are exactly the simple A-modules Ae_2/Je_2 and Ae_4/Je_4. Thus, for any $X \in A$-mod, $\mathrm{pd}_A(X) \leqslant 1$ if and only if $X \in \mathrm{add}(A \oplus (Ae_2/Je_2) \oplus (Ae_4/Je_4))$. This implies that if $\mathrm{pd}_A(X) \leqslant 1$, then the socle $\mathrm{soc}(X)$ of ${}_AX$ is a direct sum of finitely many copies of Ae_2/Je_2 and Ae_4/Je_4. Let

$$X \in \{Ae_1/Je_1,\ Ae_3/Je_3,\ Ae_2/J^2e_2,\ Ae_4/J^2e_4\}.$$

Then $\mathrm{soc}(X)$ is a direct sum of finitely many copies of simple A-modules Ae_1/Je_1 and Ae_3/Je_3. It follows that X can't be embedded into any A-module of projective dimension at most 1. By Corollary 2.3(2), $\mathrm{qpd}_A(X) \geqslant 2$. To show $\mathrm{qpd}_A(X) \leqslant 2$, we construct the following two chain maps of complexes of A-modules:

$$\begin{tikzcd}
\cdots \arrow[r, "\cdot\delta\gamma"] & Ae_3 \arrow[r, "\cdot\beta\alpha"] \arrow[d, equal] & Ae_1 \arrow[r, "\cdot\delta\gamma"] \arrow[d, equal] & Ae_3 \arrow[r, "\cdot\beta\alpha"] \arrow[d, "\cdot\beta"] & Ae_1 \arrow[r, "\cdot\delta\gamma\beta"] \arrow[d, equal] & Ae_2 \arrow[r, "\cdot\alpha"] \arrow[d] & Ae_1 \arrow[r] & 0 \\
\cdots \arrow[r, "\cdot\delta\gamma"'] & Ae_3 \arrow[r, "\cdot\beta\alpha"'] & Ae_1 \arrow[r, "\cdot\delta\gamma\beta"'] & Ae_2 \arrow[r, "\cdot\alpha"'] & Ae_1 \arrow[r] & 0, & &
\end{tikzcd}$$

where each row is a deleted projective resolution of Ae_1/Je_1, and

$$\begin{tikzcd}
\cdots \arrow[r, "\cdot\beta\alpha"] & Ae_1 \arrow[r, "\cdot\delta\gamma"] \arrow[d, equal] & Ae_3 \arrow[r, "\cdot\beta\alpha"] \arrow[d, equal] & Ae_1 \arrow[r, "\cdot\delta\gamma"] \arrow[d, "\cdot\delta"] & Ae_3 \arrow[r, "\cdot\beta\alpha\delta"] \arrow[d, "\cdot\beta"] & Ae_4 \arrow[r, "\cdot\gamma\beta"] \arrow[d] & Ae_2 \arrow[r] & 0 \\
\cdots \arrow[r, "\cdot\beta\alpha"'] & Ae_1 \arrow[r, "\cdot\delta\gamma"'] & Ae_3 \arrow[r, "\cdot\beta\alpha\delta"'] & Ae_4 \arrow[r, "\cdot\gamma\beta"'] & Ae_2 \arrow[r] & 0, & &
\end{tikzcd}$$

where each row is a deleted projective resolution of Ae_2/J^2e_2. By Theorem 1.2(4), $\mathrm{qpd}_A\,(Ae_1/Je_1) \leqslant 2$ and $\mathrm{qpd}_A(Ae_2/J^2e_2) \leqslant 2$. Thus

$$\mathrm{qpd}_A\,(Ae_1/Je_1) = \mathrm{qpd}_A(Ae_2/J^2e_2) = 2.$$

By the isomorphism Φ, we have $\mathrm{qpd}_A(Ae_3/Je_3) = 2 = \mathrm{qpd}_A(Ae_4/J^2e_4)$.

Finally, we consider the following two exact sequences:

$$0 \to Ae_1/J^2e_1 \to Ae_2 \to Ae_2/J^3e_2 \to 0, \quad 0 \to Ae_2/J^3e_2 \to Ae_4 \to Ae_4/J^2e_4 \to 0.$$

Since Ae_1/J^2e_1 is periodic, it follows from Theorem 1.1(3) that $\mathrm{qpd}_A(Ae_1/J^2e_1) = 0$. Since Ae_2 is projective-injective, $\mathrm{qpd}_A(Ae_2/J^3e_2) \leqslant 1$ by Theorem 1.2(3). If $\mathrm{qpd}_A(Ae_2/J^3e_2) = 0$, then Theorem 1.2(3) implies $\mathrm{qpd}_A(Ae_4/J^2e_4) \leqslant 1$, which contradicts that $\mathrm{qpd}_A(Ae_4/J^2e_4) = 2$. Thus $\mathrm{qpd}_A(Ae_2/J^3e_2) = 1$. Similarly, $\mathrm{qpd}_A(Ae_4/J^3e_4) = 1$ by the isomorphism Φ.

It is easy to see that all the A-modules listed in (1) and (2) have infinite projective dimension.

4. Quasi-global dimension of rings

In this section, we introduce the notion of quasi-global dimension for left Noetherian rings by analogy with the classical global dimension. In Section 4.1, we investigate some basic properties of quasi-global dimension. For instance, quasi-global dimension is equal to finitistic dimension whenever the former is finite (Proposition 4.2(1)), and both stable equivalence of Morita type and derived equivalence preserve quasi-global dimension of self-injective algebras (Corollary 4.6). In Section 4.2, we calculate the quasi-global dimension for a class of Nakayama algebras (Theorem 4.8). It turns out that these algebras have finite quasi-global dimension.

4.1. Left Noetherian rings

Throughout this section, let R be a *left Noetherian* ring. Recall that the *global* and *finitistic dimension* of R, denoted by $\mathrm{gldim}(R)$ and $\mathrm{findim}(R)$, respectively, are defined as

$$\begin{aligned}&\mathrm{gldim}(R) \coloneqq \sup\{\mathrm{pd}_R\, M \mid M \in R\text{-mod}\} \ \text{ and}\\ &\mathrm{findim}(R) \coloneqq \sup\{\mathrm{pd}_R\, M \mid M \in R\text{-mod with } \mathrm{pd}_R(M) < \infty\}.\end{aligned}$$

Related to finitistic dimension is the famous *finitistic dimension conjecture* which says that every Artin algebra has finite finitistic dimension. This conjecture is a significant open problem in the representation theory of Artin algebras.

By definition, $\mathrm{findim}(R) \leqslant \mathrm{gldim}(R)$. Now, we introduce the quasi-global dimension of R as follows.

Definition 4.1. The quasi-global dimension of R, denoted by $\mathrm{qgldim}(R)$, is defined as

$$\mathrm{qgldim}(R) \coloneqq \sup\{\mathrm{qpd}_R(M) \mid M \in R\text{-mod}\}.$$

Since $\mathrm{qpd}_R(M) \leqslant \mathrm{pd}_R(M)$ for any R-module M, we have $\mathrm{qgldim}(R) \leqslant \mathrm{gldim}(R)$. The inequality may be strict (for example, see Theorem 4.8). However, if $\mathrm{gldim}(R) < \infty$, then $\mathrm{qgldim}(R) = \mathrm{gldim}(R)$ by Theorem 1.2(1).

We first collect some basic properties of quasi-global dimension. In particular, a consequence of Proposition 4.2(1) is that the finitistic dimension conjecture holds for every Artin algebra with finite quasi-global dimension.

Proposition 4.2. *Let R and S be left Noetherian rings. Then:*

(1) $\operatorname{findim}(R) = \sup\{\operatorname{qpd}_R M \mid M \in R\text{-mod}$ *and* $\operatorname{qpd}_R M < \infty\}$. *In particular, if* $\operatorname{qgldim}(R) < \infty$, *then* $\operatorname{findim}(R) = \operatorname{qgldim}(R)$.

(2) *If* R *and* S *are Morita equivalent, then* $\operatorname{qgldim}(R) = \operatorname{qgldim}(S)$.

(3) $\operatorname{qgldim}(R \times S) = \max\{\operatorname{qgldim}(R), \operatorname{qgldim}(S)\}$.

Furthermore, if R *and* S *are finite-dimensional algebras over a field* K, *then*

(4) $\operatorname{qgldim}(R \otimes_K S) \geqslant \max\{\operatorname{qgldim}(R), \operatorname{qgldim}(S)\}$.

Proof. (1) follows from Theorem 1.2(1) and Proposition 2.2(a).

(2) Suppose that R and S are Morita equivalent. Then there exists an S-R-bimodule P such that both ${}_SP$ and P_R are finitely generated and projective and the tensor functor $P \otimes_R - : R\text{-Mod} \to S\text{-Mod}$ is an equivalence of abelian categories. Since R and S are left Noetherian, this equivalence can be restricted to an equivalence $F : R\text{-mod} \to S\text{-mod}$ of abelian categories. Let M be a finitely generated R-module with a finite quasi-projective resolution $P_\bullet$. By Corollary 2.3(1), we may assume that $P_\bullet$ belongs to $\mathscr{K}^{\mathrm{b}}(R\text{-proj})$. Since F preserves projective modules and commutes with homology functors, it follows that $F(P_\bullet)$ is a finite quasi-projective resolution of $F(M)$. Now, it is easy to show $\operatorname{qgldim}(R) = \operatorname{qgldim}(S)$.

(3) Without loss of generality, assume $\operatorname{qgldim}(R) \geqslant \operatorname{qgldim}(S)$. We first show the equality $\operatorname{qpd}_R M = \operatorname{qpd}_{R\times S}(M \times S)$ for any $M \in R\text{-mod}$. Let $P_\bullet^M$ be a quasi-projective resolution of M. By definition, $H_i(P_\bullet^M) \simeq M^{a_i}$ for some $a_i \in \mathbb{N}$. Define $S_\bullet := \bigoplus_{i\in\mathbb{Z}} S^{a_i}[a_i]$. Then $P_\bullet^M \times S_\bullet$ is a quasi-projective resolution of $M \times S$. This implies $\operatorname{qpd}_{R\times S}(M \times S) \leqslant \operatorname{qpd}_R M$. The converse $\operatorname{qpd}_R M \leqslant \operatorname{qpd}_{R\times S}(M \times S)$ follows from the fact that any quasi-projective resolution of the $(R \times S)$-module $M \times S$ is of the form $Q_\bullet^M \times Q_\bullet^S$, where $Q_\bullet^M$ and $Q_\bullet^S$ are quasi-projective resolutions of M and S, respectively. Thus $\operatorname{qpd}_R M = \operatorname{qpd}_{R\times S}(M \times S)$. It follows that $\operatorname{qgldim}(R \times S) \geqslant \max\{\operatorname{qgldim}(R), \operatorname{qgldim}(S)\}$. In particular, if $\operatorname{qgldim}(R) = \infty$, then $\operatorname{qgldim}(R\times S) = \infty$.

It remains to show that $\operatorname{qgldim}(R \times S) \leqslant \max\{\operatorname{qgldim}(R), \operatorname{qgldim}(S)\}$. Suppose $\operatorname{qgldim}(R) < \infty$. Then $\operatorname{qgldim}(S) < \infty$. It follows that finitely generated R-modules and finitely generated S-modules have finite quasi-projective resolutions. Let $M \times N$ be an arbitrary finitely generated $(R \times S)$-module. Then $M \in R\text{-mod}$ and $N \in S\text{-mod}$. We take $P_\bullet$ and $Q_\bullet$ to be any finite quasi-projective resolutions of M and N, respectively. Then $H_i(P_\bullet) \simeq M^{c_i}$ and $H_i(Q_\bullet) \simeq N^{d_i}$ for some $c_i,\, d_i \in \mathbb{N}$. Since $P_\bullet$ and $Q_\bullet$ are bounded complexes, both c_i and d_i are nonzero for finitely many i. Set

$$F_\bullet := \Big(\bigoplus_{j\in\mathbb{Z}}(P_\bullet[j])^{d_j}\Big) \times \Big(\bigoplus_{i\in\mathbb{Z}}(Q_\bullet[i])^{c_i}\Big).$$

Then $F_\bullet$ is a finite quasi-projective resolution of $M \times N$. In fact,

$$H_k(F_\bullet) = (M \times N)^{\sum_{i+j=k} c_i d_j}$$

for each $k \in \mathbb{Z}$. Note that $\sup F_\bullet = \max\{\sup P_\bullet + \operatorname{hsup} Q_\bullet, \sup Q_\bullet + \operatorname{hsup} P_\bullet\}$ and $\operatorname{hsup} F_\bullet = \operatorname{hsup} P_\bullet + \operatorname{hsup} Q_\bullet$. This implies $\operatorname{qpd}_{R\times S}(M \times N) \leqslant \max\{\operatorname{qpd}_R M, \operatorname{qpd}_S N\}$. Thus $\operatorname{qgldim}(R \times S) \leqslant \max\{\operatorname{qgldim}(R), \operatorname{qgldim}(S)\}$.

(4) The inequality holds trivially if $\operatorname{qgldim}(R \otimes_K S) = \infty$. Assume $\operatorname{qgldim}(R \otimes_K S) < \infty$. Let $M \in R$-mod. Then $M \otimes_K S \in (R \otimes_K S)$-mod. Let $P_\bullet \in \mathscr{C}^b((R \otimes_K S)\text{-proj})$ be any finite quasi-projective resolution of $M \otimes_K S$ with $H_i(P_\bullet) \simeq (M \otimes_K S)^{a_i}$ for each i, where $a_i \in \mathbb{N}$ are not all zero. Since S is a finite-dimensional K-algebra, each P_i as an R-module is finitely generated and projective, and $H_i(P_\bullet) \simeq M^{a_i \dim_K(S)}$ as R-modules. It follows that $P_\bullet \in \mathscr{C}^b(R\text{-proj})$ is a finite quasi-projective resolution of ${}_RM$. Thus $\operatorname{qpd}_R(M) \leqslant \operatorname{qgldim}_R(M \otimes_K S)$, which forces $\operatorname{qgldim}(R) \leqslant \operatorname{qgldim}(R \otimes_K S)$. Similarly, $\operatorname{qgldim}(S) \leqslant \operatorname{qgldim}(R \otimes_K S)$. $\square$

Corollary 4.3. *Let R be a quasi-Frobenius ring. Then $\operatorname{qgldim}(R) = 0$ or ∞. If $\operatorname{qgldim}(R) = 0$, then every R-module M with $\operatorname{Ext}^i_R(M, M) = 0$ for $i \geqslant 2$ is projective.*

Proof. Since R is quasi-Frobenius, each R-module of finite projective dimension is projective. This implies $\operatorname{findim}(R) = 0$. By Proposition 4.2(1), $\operatorname{qgldim}(R) = 0$ or ∞. If $\operatorname{qgldim}(R) = 0$, then each R-module M has quasi-projective dimension 0. If, in addition, $\operatorname{Ext}^i_R(M, M) = 0$ for all $i \geqslant 2$, then M is projective by Corollary 3.6. $\square$

Remark 4.4. Let R be a self-injective Artin algebra. Recall that *Tachikawa's second conjecture* asserts that a finitely generated R-module M is projective if $\operatorname{Ext}^i_R(M, M) = 0$ for $i \geqslant 1$. By Corollary 4.3, Tachikawa's second conjecture holds for a self-injective Artin algebra with finite quasi-global dimension (or equivalently, with quasi-global dimension 0). Note that if R is of *finite representation type*, that is, there are only finitely many isomorphism classes of indecomposable, finitely generated R-modules, then $\operatorname{qgldim}(R) = 0$. In this case, each $M \in R$-mod is periodic and $\operatorname{qpd}_R(M) = 0$ by Theorem 1.1(3).

It would be interesting to classify all self-injective Artin algebras with quasi-global dimension 0.

Proposition 4.5. *Let A and B be quasi-Frobenius rings. Suppose that $F : A$-mod $\to B$-mod is an exact functor and induces an equivalence A-$\underline{\text{mod}} \to B$-$\underline{\text{mod}}$ of stable module categories. Then $\operatorname{qgldim}(B) \leqslant \operatorname{qgldim}(A)$. In particular, if $\operatorname{qgldim}(A) = 0$, then $\operatorname{qgldim}(B) = 0$.*

Proof. Since F is an exact functor and induces an equivalence A-$\underline{\text{mod}} \to B$-$\underline{\text{mod}}$, we see that F commutes with homology functors, and preserves and detects finitely generated projective modules. It follows that if $P_\bullet := (P_i)_{i\in\mathbb{Z}}$ is a quasi-projective resolution of an A-module M, then $F(P_\bullet) := \big(F(P_i)_{i\in\mathbb{Z}}\big)$ is a quasi-projective resolution of the B-module $F(M)$. This implies $\operatorname{qpd}_B(F(M)) \leqslant \operatorname{qpd}_A(M)$. Let $Y \in B$-mod. As $F : A$-$\underline{\text{mod}} \to B$-$\underline{\text{mod}}$

is dense, there exists $X \in A$-mod and $P_1, P_2 \in B$-proj such that $Y \oplus P_1 \simeq F(X) \oplus P_2$. Since A and B are quasi-Frobenius, $\mathrm{qpd}_B(Y) = \mathrm{qpd}_B(F(X))$ by Corollary 3.9. Thus $\mathrm{qpd}_B(Y) \leqslant \mathrm{qpd}_A(X)$. This forces $\mathrm{qgldim}(B) \leqslant \mathrm{qgldim}(A)$. $\square$

Next, we apply Proposition 4.5 to special stable equivalences between self-injective algebras. For the definitions and constructions of stable equivalences of Morita type and derived equivalences, we refer to the survey article [14].

Corollary 4.6. *Let A and B be finite-dimensional self-injective algebras over a field K. If A and B are stably equivalent of Morita type or derived equivalent, then they have the same quasi-global dimension.*

Proof. Note that finite-dimensional self-injective algebras are quasi-Frobenius rings. If A and B are stably equivalent of Morita type, then there exists an A-B-module X and a B-A-module Y such that the tensor functors $Y \otimes_A - : A\text{-mod} \to B\text{-mod}$ and $X \otimes_B - : B\text{-mod} \to A\text{-mod}$ are exact and induce equivalences $A\text{-}\underline{\mathrm{mod}} \simeq B\text{-}\underline{\mathrm{mod}}$ of stable module categories. In this case, $\mathrm{qgldim}(B) = \mathrm{qgldim}(A)$ by Proposition 4.5. If A and B are derived equivalent, then they are stably equivalent of Morita type by [11, Corollary 5.5] and thus $\mathrm{qgldim}(B) = \mathrm{qgldim}(A)$. $\square$

Finally, we provide an example of a self-injective algebra with *infinite* quasi-global dimension.

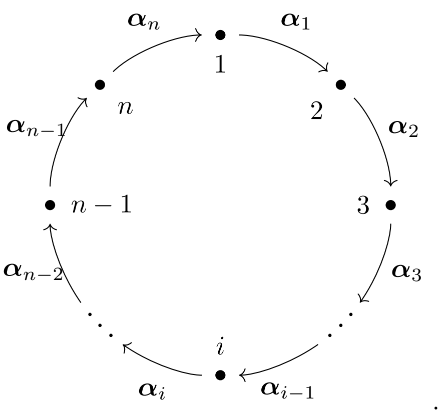
Let K be a field with a non-zero element $q \in K$ that is not a root of unity. The local symmetric K-algebra A, defined by Liu–Schulz (see [10]), is generated by x_0, x_1, x_2 with the relations: $x_i^2 = 0$, and $x_{i+1}x_i + qx_ix_{i+1} = 0$ for $i = 0, 1, 2$.

Proposition 4.7. *Let A be Liu–Schulz algebra. Then* $\mathrm{qgldim}(A) = \infty$.

Proof. Since A is a finite-dimensional symmetric K-algebra, every non-projective module has infinite projective dimension. By [5, Lemma 6.6], there exists a finitely generated non-projective A-module I_0 such that $\mathrm{Ext}_A^i(I_0, I_0) = 0$ for all $i \geqslant 2$. It follows from Theorem 1.2(2) that $\mathrm{qpd}_A(I_0) = \infty$. Thus $\mathrm{qgldim}(A) = \infty$. $\square$

4.2. Nakayama algebras

In this subsection, we fix an arbitrary field K, an integer $n \geqslant 2$ and the following quiver Q:



Denote by

$$\rho_i = \begin{cases} \alpha_n \cdots \alpha_1 & \text{for } i = 1, \\ \alpha_{i-1} \cdots \alpha_1 \alpha_n \cdots \alpha_i & \text{for } 2 \leqslant i \leqslant n, \end{cases}$$

the composition of certain arrows in Q. Let m and λ_j be integers with $1 \leqslant m \leqslant n$, $1 \leqslant j \leqslant m$ and $1 \leqslant \lambda_1 < \lambda_2 < \cdots < \lambda_m \leqslant n$. We define

$$A_{n,m} \coloneqq KQ/\langle \rho_{\lambda_1}, \rho_{\lambda_2}, \ldots, \rho_{\lambda_m} \rangle,$$

the quotient algebra of the path algebra KQ modulo the ideal generated by ρ_{λ_i} for all $1 \leqslant i \leqslant m$. Clearly $A_{n,m}$ is a Nakayama algebra. The main result of this subsection is to compare its quasi-global dimension with its global dimension.

Theorem 4.8. (1) $\mathrm{gldim}(A_{n,1}) = \mathrm{qgldim}(A_{n,1}) = 2$.

(2) $\mathrm{gldim}(A_{n,n}) = \infty$ *and* $\mathrm{qgldim}(A_{n,n}) = 0$.

(3) *If* $n > 2$ *and* $1 < m < n$, *then* $\mathrm{gldim}(A_{n,m}) = \infty$ *and* $\mathrm{qgldim}(A_{n,m}) = 2$.

To show Theorem 4.8, we introduce some notation and establish several lemmas.

Let J be the Jacobson radical of the algebra $A_{n,m}$. The Loewy length of an $A_{n,m}$-module M is denote by

$$\ell(M) \coloneqq \min\{s \in \mathbb{N}^+ \mid J^s M = 0\}.$$

Let S_i and P_i be the simple and indecomposable projective $A_{n,m}$-module corresponding to each vertex $i \in Q_0$, respectively. For simplicity, we denote by $L(i,k)$ the indecomposable $A_{n,m}$-module that has top S_i and Loewy length k. In particular, $L(i,1) = S_i$. Moreover, we set $\Delta \coloneqq \{\lambda_1, \cdots \lambda_m\}$ and define a bijection $\sigma : \{1, \cdots, n\} \to \{1, \cdots, n\}$ by

$$\sigma(i) = \begin{cases} i+1 & \text{for } 1 \leqslant i \leqslant n-1, \\ 1 & \text{for } i = n. \end{cases}$$

For $r \geqslant 1$, we denote the compositions $\underbrace{\sigma \cdots \sigma}_{r \text{ times}}$ and $\underbrace{\sigma^{-1} \cdots \sigma^{-1}}_{r \text{ times}}$ by σ^r and σ^{-r}, respectively. As usual, σ^0 stands for the identity map of $\{1, \cdots, n\}$. In the following, we always fix $i \in \{1, \ldots, n\}$.

Lemma 4.9. *The following statements are true.*

(1) *For any $A_{n,m}$-module $L(i,k)$, we have $0 < k < 2n$.*

(2) *$\ell(P_i) = n + \inf\{r \in \mathbb{N} \mid \sigma^r(i) \in \Delta\}$. In particular, $n \leqslant \ell(P_i) < 2n$.*

(3) *If $n \leqslant k < \ell(P_i)$, then $\ell(P_{\sigma^k(i)}) = \ell(P_i) - (k-n)$.*

(4) *If $0 < k < \ell(P_i)$, then $\Omega(L(i,k)) \simeq L\big(\sigma^k(i), \ell(P_i) - k\big)$.*

Proof. (1) Clearly $k > 0$. On the other hand, assume $k \geqslant 2n$. Since $L(i,k)$ is a quotient of P_i, we have $\ell(P_i) \geqslant k \geqslant 2n$. This implies that the path $p_i := \alpha_{\sigma^{n-2}(i)} \cdots \alpha_{\sigma(i)} \alpha_i \alpha_{\sigma^{n-1}(i)} \cdots \alpha_{\sigma(i)} \alpha_i$ is nonzero in $A_{n,m}$. In the quiver Q, any path of length n (for example, ρ_j for $1 \leqslant j \leqslant n$) is a subpath of p_i. Thus $\rho_j \neq 0$ in $A_{n,m}$, which is a contradiction to the definition of $A_{n,m}$.

(2) By definition of $A_{n,m}$, we have $\ell(P_i) \geqslant n$. Clearly $\ell(P_i) = n$ if and only if $i \in \Delta$. Suppose $\ell(P_i) = n + r$ for some $r > 0$. Then

$$\alpha_{\sigma^{n+r-2}(i)} \cdots \alpha_{\sigma(i)} \alpha_i \neq 0 \quad \text{and} \quad \alpha_{\sigma^{n+r-1}(i)} \alpha_{\sigma^{n+r-2}(i)} \cdots \alpha_{\sigma(i)} \alpha_i = 0.$$

This implies

$$\alpha_{\sigma^{n+r-2}(i)} \cdots \alpha_{\sigma^r(i)} \neq 0 \quad \text{and} \quad \alpha_{\sigma^{n+r-1}(i)} \alpha_{\sigma^{n+r-2}(i)} \cdots \alpha_{\sigma^r(i)} = 0.$$

Thus $\sigma^r(i) \in \Delta$.

We now prove that if there exists an integer $t > 0$ such that $\sigma^t(i) \in \Delta$, then $t \geqslant r$.

In fact, if $t < r$, then $\alpha_{\sigma^{n+r-2}(i)} \cdots \alpha_{\sigma^r(i)} \cdots \alpha_{\sigma^t(i)} \neq 0$, and therefore $n+r-2-t+1 = n + (r-t) - 1 \geqslant n$, which contradicts $\sigma^t(i) \in \Delta$.

(3) Assume $\ell(P_i) = n+r$ with $r \in \mathbb{N}^+$. Then $\alpha_{\sigma^{n+r-2}(i)} \cdots \alpha_{\sigma(i)} \alpha_i \neq 0$. By (1), $r < n$. By (2), $\sigma^r(i) \in \Delta$. It follows that

$$\alpha_{\sigma^{n+r-1}(i)} \alpha_{\sigma^{n+r-2}(i)} \cdots \alpha_{\sigma^n(i)} \cdots \alpha_{\sigma^r(i)} = 0.$$

The case $k = n$ is clear. Now assume $k = n + s$ with $s > 0$. Then $s < r$. This implies

$$\alpha_{\sigma^{n+r-2}(i)} \cdots \alpha_{\sigma^r(i)} \cdots \alpha_{\sigma^s(i)} \neq 0 \quad \text{and} \quad \alpha_{\sigma^{n+r-1}(i)} \alpha_{\sigma^{n+r-2}(i)} \cdots \alpha_{\sigma^r(i)} \cdots \alpha_{\sigma^s(i)} = 0.$$

Thus (3) holds.

(4) Note that $\Omega(L(i,k)) = J^k e_i$. Let $f\colon P_{\sigma^k(i)} \to J^k e_i$ be the $A_{n,m}$-module homomorphism induced by $e_{\sigma^k(i)} \mapsto \alpha_{\sigma^{k-1}(i)}\alpha_{\sigma^{k-2}(i)}\cdots\alpha_i$. Then $\mathrm{Ker} f = J^{\ell(P_i)-k}e_{\sigma^k(i)}$ and thus (4) holds. $\square$

Lemma 4.10. *Let $n \leqslant k < \ell(P_i)$. Then* $\mathrm{qpd}_{A_{n,m}}(L(i,k)) = \mathrm{pd}_{A_{n,m}}(L(i,k)) = 2$.

Proof. Let $\ell(P_i) = n + r$ with $r > 0$. By Lemma 4.9(2), $i \notin \Delta$. Since $k < \ell(P_i)$, we see from Lemma 4.9(4) that $\Omega(L(i,k)) \simeq L(\sigma^k(i), n + r - k)$. By $k \geqslant n$ and Lemma 4.9(3), $\ell(P_{\sigma^k(i)}) = n + r - (k - n) = 2n + r - k$. Since $2n + r - k > n + r - k$, it is clear that $L(\sigma^k(i), n + r - k)$ is non-projective. Thus $\Omega(L(i,k))$ is non-projective. Now, there is a series of isomorphisms:

$$\Omega^2(L(i,k)) \simeq \Omega(L(\sigma^k(i), n{+}r{-}k)) \simeq L(\sigma^{n+r}(i), (2n{+}r{-}k){-}(n{+}r{-}k)) = L(\sigma^{n+r}(i), n).$$

Note that $\sigma^{n+r}(i) = \sigma^r(i)$. By Lemma 4.9(2), $\sigma^r(i) \in \Delta$, which forces $\ell(P_{\sigma^{n+r}(i)}) = n$. This implies that $\Omega^2(L(i,k))$ is projective, and therefore $\mathrm{pd}_{A_{n,m}}(L(i,k)) = 2$. By Proposition 3.1, $\mathrm{qpd}_{A_{n,m}}(L(i,k)) = \mathrm{pd}_{A_{n,m}}(L(i,k))$. $\square$

Lemma 4.11. *Let $0 < k < n$. Then*

$$\mathrm{qpd}_{A_{n,m}}(L(i,k)) \leqslant \begin{cases} 0 & \textit{if } i,\ \sigma^k(i) \in \Delta; \\ 1 & \textit{if } i \notin \Delta,\ \sigma^k(i) \in \Delta; \\ 2 & \textit{if } \sigma^k(i) \notin \Delta. \end{cases}$$

Proof. Since $0 < k < n$, we have $\Omega(L(i,k)) \simeq L\big(\sigma^k(i), \ell(P_i) - k\big)$ by Lemma 4.9(2)(4). In the following, we divide the proof of Lemma 4.11 into four cases.

Case 1. i, $\sigma^k(i) \in \Delta$.

By Lemma 4.9(2), $\ell(P_i) = \ell(P_{\sigma^k(i)}) = n$ and $L\big(\sigma^k(i), \ell(P_i) - k\big) = L(\sigma^k(i), n - k)$ which is non-projective. It follows from Lemma 4.9(4) that

$$\Omega^2(L(i,k)) \simeq \Omega(L(\sigma^k(i), n - k)) \simeq L(\sigma^n(i), n - (n - k)) = L(i,k).$$

Thus $L(i,k)$ is periodic. By Theorem 1.1(3), $\mathrm{qpd}_{A_{n,m}}(L(i,k)) = 0$.

In the later proof, we will use the following (minimal) projective resolution of the periodic module $L(i,k)$ for several times:

$$\cdots \longrightarrow P_{\sigma^k(i)} \xrightarrow{\ g\ } P_i \xrightarrow{\ h\ } P_{\sigma^k(i)} \xrightarrow{\ g\ } P_i \longrightarrow L(i,k) \longrightarrow 0 \qquad (*)$$

where g and h are induced by

$$e_{\sigma^k(i)} \mapsto \alpha_{\sigma^{k-1}(i)}\cdots\alpha_{\sigma(i)}\alpha_i \quad \text{and} \quad e_i \mapsto \alpha_{\sigma^{n-1}(i)}\cdots\alpha_{\sigma^{k+1}(i)}\alpha_{\sigma^k(i)},$$

respectively.

Case 2. $i \notin \Delta$ and $\sigma^k(i) \in \Delta$.

By Lemma 4.9(2), $\ell(P_i) = n + r$ for some $r \in \mathbb{N}^+$ such that $\sigma^r(i) \in \Delta$ and $k \geqslant r$.

If $k = r$, then $L(\sigma^k(i), n + r - k) = L(\sigma^r(i), n)$ which is projective by Lemma 4.9(2). In this case, $\mathrm{qpd}_{A_{n,m}}(L(i,k)) = \mathrm{pd}_{A_{n,m}}(L(i,k)) = 1$, due to Proposition 3.1.

Suppose $k > r$. Then $\sigma^k(i) \in \Delta$ and $\sigma^{n+r-k}(\sigma^k(i)) = \sigma^{n+r}(i) = \sigma^r(i) \in \Delta$. By Case 1, $L(\sigma^k(i), n+r-k)$ is periodic, and so is $\Omega(L(i,k))$. Applying $(*)$ to $\Omega(L(i,k))$, we can construct a projective resolution of $L(i,k)$ as follows:

$$\cdots \xrightarrow{g_1} P_{\sigma^k(i)} \xrightarrow{h_1} P_{\sigma^r(i)} \xrightarrow{g_1} P_{\sigma^k(i)} \xrightarrow{f} P_i \longrightarrow L(i,k) \longrightarrow 0$$

where f, g_1 and h_1 are induced by

$$e_{\sigma^k(i)} \mapsto \alpha_{\sigma^{k-1}(i)} \cdots \alpha_{\sigma(i)}\alpha_i, \quad e_{\sigma^r(i)} \mapsto \alpha_{\sigma^{n+r-1}(i)} \cdots \alpha_{\sigma^{k+1}(i)}\alpha_{\sigma^k(i)},$$
$$e_{\sigma^k(i)} \mapsto \alpha_{\sigma^{k-1}(i)} \cdots \alpha_{\sigma^{r+1}(i)}\alpha_{\sigma^r(i)},$$

respectively. Let $\beta : P_{\sigma^r(i)} \to P_i$ be the $A_{n,m}$-module homomorphism induced by $e_{\sigma^r(i)} \mapsto \alpha_{\sigma^{r-1}(i)} \cdots \alpha_{\sigma(i)}\alpha_i$. Then we obtain the following commutative diagram:

$$\begin{tikzcd}
\cdots \arrow[r, "h_1"] & P_{\sigma^r(i)} \arrow[r, "g_1"] \arrow[d, equal] & P_{\sigma^k(i)} \arrow[r, "h_1"] \arrow[d, equal] & P_{\sigma^r(i)} \arrow[r, "g_1"] \arrow[d, "\beta"] & P_{\sigma^k(i)} \arrow[r, "f"] \arrow[d] & P_i \arrow[r] \arrow[d] & 0 \arrow[d] \\
\cdots \arrow[r, "h_1"] & P_{\sigma^r(i)} \arrow[r, "g_1"] & P_{\sigma^k(i)} \arrow[r, "f"] & P_i \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & 0.
\end{tikzcd}$$

By Theorem 1.2(4), $\mathrm{qpd}_{A_{n,m}}(L(i,k)) \leqslant 1$.

Case 3. $i \in \Delta$ and $\sigma^k(i) \notin \Delta$.

By Lemma 4.9(2), $\ell(P_i) = n$ and $\ell(P_{\sigma^k(i)}) = n + r$ for some $r > 0$ with $\sigma^r(\sigma^k(i)) = \sigma^{r+k}(i) \in \Delta$. Then $\Omega(L(i,k)) \simeq L\big(\sigma^k(i), \ell(P_i) - k\big) = L(\sigma^k(i), n-k)$ which is non-projective by $n - k < n$. It follows that

$$\Omega^2(L(i,k)) \simeq \Omega(L(\sigma^k(i), n-k)) \simeq L(\sigma^n(i), r+k) = L(i, r+k).$$

If $L(i, r+k)$ is projective, then $\mathrm{qpd}_{A_{n,m}}(L(i,k)) = \mathrm{pd}_{A_{n,m}}(L(i,k)) = 2$. Now, assume that $L(i, r+k)$ is non-projective. Since $\ell(P_i) = n$ and $L(i, r+k)$ is a quotient of P_i, we have $0 < r + k < n$. Recall that $i \in \Delta$ and $\sigma^{r+k}(i) \in \Delta$. By Case 1, the module $L(i, r+k)$ is periodic. Similarly, by $(*)$, we can construct a projective resolution of $L(i,k)$ as follows:

$$\cdots \xrightarrow{g_2} P_i \xrightarrow{h_2} P_{\sigma^{r+k}(i)} \xrightarrow{g_2} P_i \xrightarrow{f_1} P_{\sigma^k(i)} \xrightarrow{f_0} P_i \longrightarrow L(i,k) \longrightarrow 0,$$

where f_0, f_1, g_2 and h_2 are induced by

$$e_{\sigma^k(i)} \mapsto \alpha_{\sigma^{k-1}(i)} \cdots \alpha_{\sigma(i)}\alpha_i, \quad e_i \mapsto \alpha_{\sigma^{n-1}(i)} \cdots \alpha_{\sigma^{k+1}(i)}\alpha_{\sigma^k(i)},$$

$$e_{\sigma^{r+k}(i)} \mapsto \alpha_{\sigma^{r+k-1}(i)} \cdots \alpha_{\sigma(i)}\alpha_i, \quad e_i \mapsto \alpha_{\sigma^{n-1}(i)} \cdots \alpha_{\sigma^{r+k+1}(i)}\alpha_{\sigma^{r+k}(i)},$$

respectively. Further, let $\gamma : P_{\sigma^{r+k}(i)} \to P_{\sigma^k(i)}$ be the $A_{n,m}$-module homomorphism induced by $e_{\sigma^{r+k}(i)} \mapsto \alpha_{\sigma^{r+k-1}(i)} \cdots \alpha_{\sigma^{k+1}(i)}\alpha_{\sigma^k(i)}$. Then there is a commutative diagram:

$$\begin{tikzcd}
\cdots \arrow[r, "g_2"] & P_i \arrow[r, "h_2"] \arrow[d, equal] & P_{\sigma^{r+k}(i)} \arrow[r, "g_2"] \arrow[d, "\gamma"] & P_i \arrow[r, "f_1"] \arrow[d, equal] & P_{\sigma^k(i)} \arrow[r, "f_0"] \arrow[d] & P_i \arrow[r] \arrow[d] & 0 \arrow[d] \\
\cdots \arrow[r, "g_2"] & P_i \arrow[r, "f_1"] & P_{\sigma^k(i)} \arrow[r, "f_0"] & P_i \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & 0.
\end{tikzcd}$$

By Theorem 1.2(4), $\mathrm{qpd}_{A_{n,m}}(L(i,k)) \leqslant 2$.

Case 4. $i \notin \Delta$ and $\sigma^k(i) \notin \Delta$.

By Lemma 4.9(2), $\ell(P_i) = n+r_1$ and $\ell(P_{\sigma^k(i)}) = n+r_2$ for $0 < r_1 < n$ and $0 < r_2 < n$ such that $\sigma^{r_1}(i), \sigma^{r_2+k}(i) \in \Delta$. By Lemma 4.9(4), $\Omega(L(i,k)) \simeq L(\sigma^k(i), n+r_1-k)$.

If either $\Omega(L(i,k))$ or $\Omega^2(L(i,k))$ is projective, then $\mathrm{pd}_{A_{n,m}}(L(i,k)) \leqslant 2$. In this case, $\mathrm{qpd}_{A_{n,m}}(L(i,k)) = \mathrm{pd}_{A_{n,m}}(L(i,k)) \leqslant 2$ by Proposition 3.1.

Now, assume that both $\Omega(L(i,k))$ and $\Omega^2(L(i,k))$ are non-projective. Then $n+r_1-k < \ell(P_{\sigma^k(i)}) = n+r_2$. It follows that

$$\Omega^2(L(i,k)) \simeq \Omega\big(L(\sigma^k(i), n+r_1-k)\big) \simeq L(\sigma^{n+r_1}(i), r_2-r_1+k) = L(\sigma^{r_1}(i), r_2-r_1+k).$$

By $\sigma^{r_1}(i) \in \Delta$, we have $\ell(P_{\sigma^{r_1}(i)}) = n$ by Lemma 4.9(2). Since $\Omega^2(L(i,k))$ is non-projective, it is clear that $0 < r_2 - r_1 + k < n$. Note that $\sigma^{r_1}(i) \in \Delta$ and $\sigma^{r_2-r_1+k}(\sigma^{r_1}(i)) = \sigma^{r_2+k}(i) \in \Delta$. Thus $L(\sigma^{r_1}(i), r_2-r_1+k)$ is periodic by Case 1. Still by $(*)$, we can construct a projective resolution of $L(i,k)$ as follows:

$$\cdots \xrightarrow{g_3} P_{\sigma^{r_1}(i)} \xrightarrow{h_3} P_{\sigma^{r_2+k}(i)} \xrightarrow{g_3} P_{\sigma^{r_1}(i)} \xrightarrow{d_1} P_{\sigma^k(i)} \xrightarrow{d_0} P_i \longrightarrow L(i,k) \longrightarrow 0$$

where d_0, d_1, g_3 and h_3 are induced by

$$e_{\sigma^k(i)} \mapsto \alpha_{\sigma^{k-1}(i)} \cdots \alpha_{\sigma(i)}\alpha_i, \quad e_{\sigma^{r_1}(i)} \mapsto \alpha_{\sigma^{n+r_1-1}(i)} \cdots \alpha_{\sigma^{k+1}(i)}\alpha_{\sigma^k(i)},$$
$$e_{\sigma^{r_2+k}(i)} \mapsto \alpha_{\sigma^{r_2+k-1}(i)} \cdots \alpha_{\sigma^{r_1+1}(i)}\alpha_{\sigma^{r_1}(i)},$$
$$e_{\sigma^{r_1}(i)} \mapsto \alpha_{\sigma^{n+r_1-1}(i)} \cdots \alpha_{\sigma^{r_2+k+1}(i)}\alpha_{\sigma^{r_2+k}(i)}.$$

Further, let $\delta : P_{\sigma^{r_2+k}(i)} \to P_{\sigma^k(i)}$ and $\epsilon : P_{\sigma^{r_1}(i)} \to P_i$ be the $A_{n,m}$-module homomorphisms induced by

$$e_{\sigma^{r_2+k}(i)} \mapsto \alpha_{\sigma^{r_2+k-1}(i)} \cdots \alpha_{\sigma^{k+1}(i)}\alpha_{\sigma^k(i)} \quad \text{and} \quad e_{\sigma^{r_1}(i)} \mapsto \alpha_{\sigma^{r_1-1}(i)} \cdots \alpha_{\sigma(i)}\alpha_i,$$

respectively. Then there is a commutative diagram:

$$\begin{tikzcd}
\cdots \arrow[r, "g_3"] & P_{\sigma^{r_1}(i)} \arrow[r, "h_3"] \arrow[d, equal] & P_{\sigma^{r_2+k}(i)} \arrow[r, "g_3"] \arrow[d, "\delta"] & P_{\sigma^{r_1}(i)} \arrow[r, "d_1"] \arrow[d, "\epsilon"] & P_{\sigma^k(i)} \arrow[r, "d_0"] \arrow[d] & P_i \arrow[r] \arrow[d] & 0 \arrow[d] \\
\cdots \arrow[r, "g_3"] & P_{\sigma^{r_1}(i)} \arrow[r, "d_1"] & P_{\sigma^k(i)} \arrow[r, "d_0"] & P_i \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & 0.
\end{tikzcd}$$

Thus $\mathrm{qpd}_{A_{n,m}}(L(i,k)) \leqslant 2$ by Theorem 1.2(4). $\square$

Proof of Theorem 4.8 (1)**.** Without loss of generality, we can assume $\lambda_1 = 1$. It can be checked that

$$\mathrm{pd}_{A_{n,m}}(S_i) = \begin{cases} 2, & \text{if } i = 1, \\ 1, & \text{otherwise.} \end{cases}$$

(2) $A_{n,n}$ is a self-injective algebra of finite representation type. By Remark 4.4, $\mathrm{qgldim}(A_{n,n}) = 0$.

(3) Note that $\{L(i,k) \mid 1 \leqslant i \leqslant n, 1 \leqslant k \leqslant \ell(P_i)\}$ is the set of isomorphism classes of indecomposable $A_{n,m}$-modules. By Theorem 1.1(1),

$$\mathrm{qgldim}(A_{n,m}) = \sup\{\mathrm{qpd}_{A_{n,m}}(L(i,k)) \mid 1 \leqslant i \leqslant n, 0 < k \leqslant \ell(P_i)\}.$$

Combining Lemma 4.10 with Lemma 4.11, we have $\mathrm{qpd}_{A_{n,m}}(L(i,k)) \leqslant 2$ for $1 \leqslant i \leqslant n$ and $0 < k \leqslant \ell(P_i)$. This implies $\mathrm{qgldim}(A_{n,m}) \leqslant 2$. Moreover, since $\Delta \neq \{1, 2, \cdots, n\}$, there exists an indecomposable projective $A_{n,m}$-module P_i such that $\ell(P_i) > n$ by Lemma 4.9(2). It follows from Lemma 4.10 that $\mathrm{qpd}_{A_{n,m}}(L(i,n)) = \mathrm{pd}_{A_{n,m}}(L(i,n)) = 2$. Thus $\mathrm{qgldim}(A_{n,m}) = 2$. Since $n > 2$ and $1 < m < n$, there exists an integer $i \in \Delta$ such that $\sigma^k(i) \in \Delta$ for some integer k with $0 < k < n$. According to Case 1 in the proof of Lemma 4.11, the module $L(i,k)$ is periodic but non-projective. This implies $\mathrm{gldim}(A_{n,m}) = \infty$. $\square$

Acknowledgments

The research work was supported by the National Natural Science Foundation of China (Grant 12031014). All of the authors are very grateful to anonymous referees for constructive suggestions on language and presentation of results in the manuscript.

Data availability

No data was used for the research described in the article.

References

[1] I. Assem, D. Simson, A. Skowroński, Elements of the Representation Theory of Associative Algebras, vol. 1. Techniques of Representation Theory, London Mathematical Society Student Texts, vol. 65, Cambridge University Press, Cambridge, 2006.

[2] M. Auslander, Modules over unramified regular local rings, Ill. J. Math. 5 (1961) 631–647.
[3] B. Briggs, E. Grifo, J. Pollitz, Constructing nonproxy small test modules for the complete intersection property, Nagoya Math. J. 246 (2022) 412–429.
[4] H.X. Chen, M. Fang, C.C. Xi, Tachikawa's second conjecture, derived recollements, and gendo-symmetric algebras, Compos. Math. 160 (2024) 2704–2734.
[5] H.X. Chen, S.Y. Pan, C.C. Xi, Inductions and restrictions for stable equivalences of Morita type, J. Pure Appl. Algebra 216 (2012) 643–661.
[6] H.X. Chen, C.C. Xi, Homological theory of self-orthogonal modules, Trans. Am. Math. Soc. 378 (2025) 7287–7335.
[7] M. Gheibi, Quasi-injective dimension, J. Pure Appl. Algebra 228 (2024) 107468.
[8] M. Gheibi, D.A. Jorgensen, R. Takahashi, Quasi-projective dimension, Pac. J. Math. 312 (2021) 113–147.
[9] V.H. Jorge-pérez, P. Martins, V.D. Mendoza-rubio, A generalized depth formula for modules of finite quasi-projective dimension, arXiv:2409.08996v2, 2024.
[10] S.P. Liu, R. Schulz, The existence of bounded infinite DTr-orbits, Proc. Am. Math. Soc. 122 (1994) 1003–1005.
[11] J. Rickard, Derived equivalences as derived functors, J. Lond. Math. Soc. (2) 43 (1991) 37–48.
[12] N.M. Tri, Chouinard's formula for quasi-injective dimension, Rev. R. Acad. Cienc. Exactas Fís. Nat., Ser. A Mat. 119 (2025) 52.
[13] C.A. Weibel, An Introduction to Homological Algebra, Cambridge Studies in Advanced Mathematics, vol. 38, Cambridge University Press, Cambridge, 1994.
[14] C.C. Xi, Derived equivalences of algebras, Bull. Lond. Math. Soc. 50 (2018) 945–985.